King Abdulaziz University
Faculty of Computing and information Technology
Computer Science Department

CPCS 211: Digital Logic Design

Chapter3 : The Karnaugh Map

Outline

- Introduction to K-Maps
- Implicants and Prime Implicants
- Examples
- Don’t Care
- Five and Six Variables Maps
- Multiple Output Problems

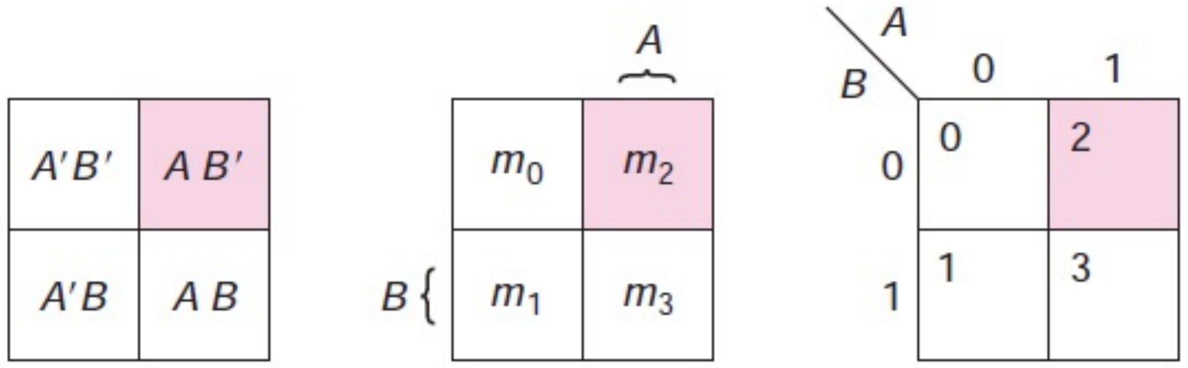
Introduction to K-Maps

- Used to simplify Boolean functions.
- Consists of one square for each possible minterm in a function.
- Thus, a two-variable map has 4 squares, a three variable map has 8 squares, and a four-variable map has 16 squares.
- K-map is a tool for simplifying combinational logic with 2, 3, 4 variables, or more.

Karnaugh Map: 2-Variables

- For 2 variables, 4 cells are required (2^2).
- The map shown is for two variables labeled A, B, each cell represents one possible product term, and each cell differs from an adjacent cell by only one variable.

Map 3.1 Two-variable Karnaugh maps.



Two Variables K-Maps

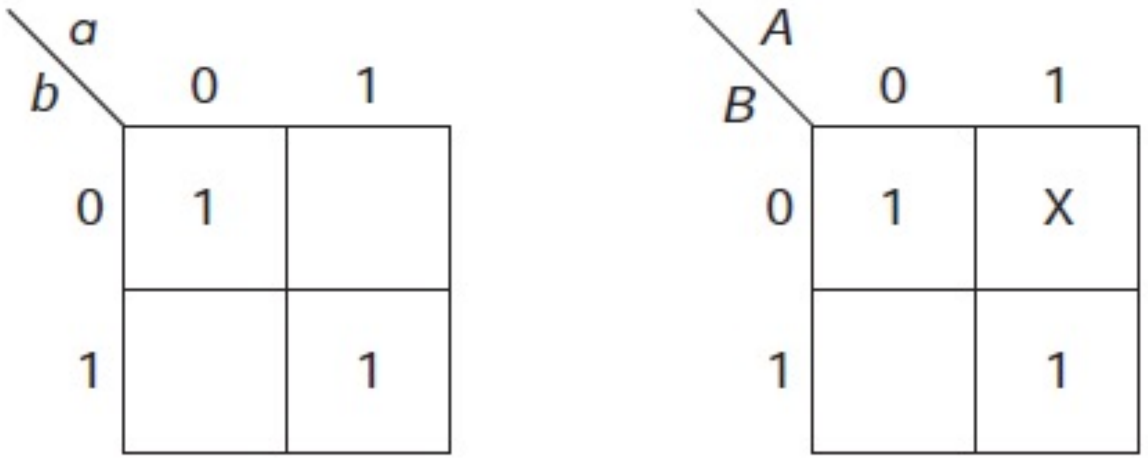
- A two-variable K-Map is shown
- When plot a function, put
  - **1** in each square corresponding to minterm that is included in function
  - **0** in or leave blank those squares not included in function
  - **X** in square for is don’t care minterm

Example

The K-Maps for the functions

$f(a, b) = \sum m(0,3)$ and $g(A, B) = \sum m(0, 3) + \sum d(2)$

Map 3.2 Plotting functions.



$f(a, b) = \Sigma m\,(0, 3)$ $g(A, B) = \Sigma m\,(0, 3) + \Sigma d\,(2)$

Three Variables K-Maps

- There are **$2^3=8$** minterms, so 3-variable K-Map consists of eight squares

Map 3.3 Three-variable maps.

C \ AB	A′B′ 00	A′B 01	AB 11	AB′ 10
C′ 0	A′B′C′	A′BC′	ABC′	AB′C′
C 1	A′B′C	A′BC	ABC	AB′C

C \ AB	00	01	11	10
0	0	2	6	4
1	1	3	7	5

- Note that minterms are arranged in sequence similar to reflection code
- Characteristic: only 1-bit changes from 1 to 0 or 0 to 1, in listing sequence

Three Variables K-Maps

- Some people label the row(s) of the map with the first variable(s) and the columns with the others

- The three-variable map then looks like

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Three Variables K-Maps

- Notice that the last two columns are not in numeric order

- That is the key idea that makes the map work. By organizing the map that way, the minterms in adjacent squares can always be combined using the adjacency property:

P9a. $ab + ab' = a$

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Product Terms

Product terms that correspond to the sum of two minterms appear as two adjacent 1's on the map

Product Terms

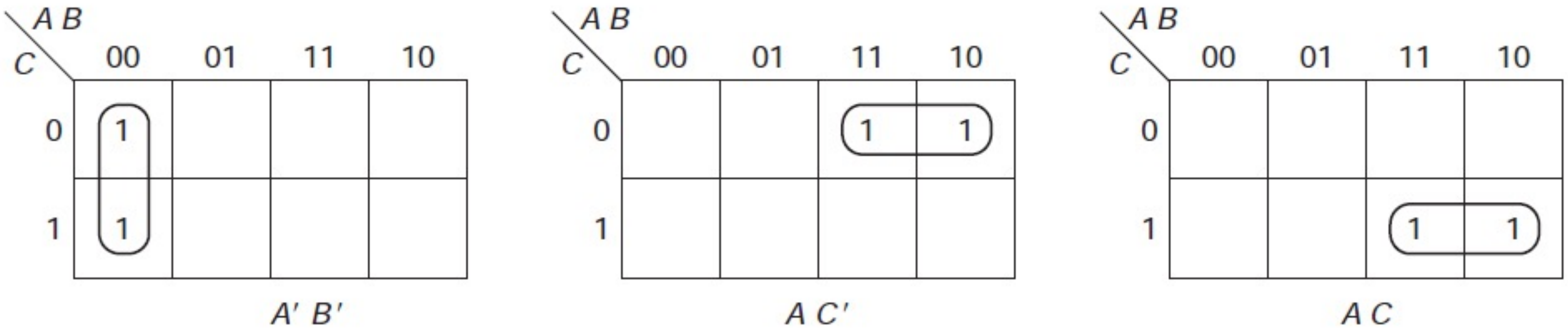
The following terms are adjacent; shown in Map 3.5:

– *m0 + m1: A'B'C' + A'B'C = A'B'*

– *m4 +m6:AB'C' +ABC' =AC'*

– *m7 +m5:ABC +AB'C =AC*

Map 3.5 Product terms corresponding to groups of two.



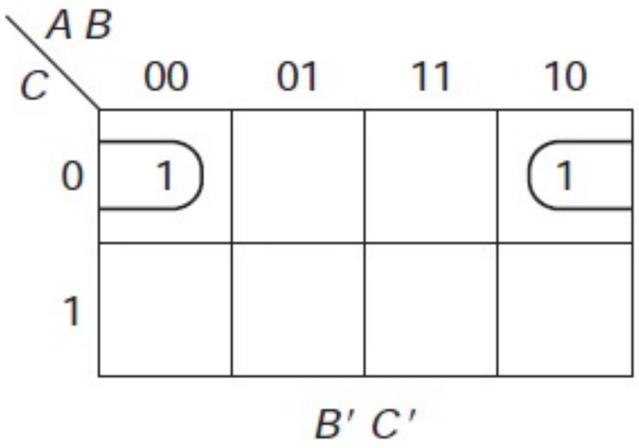
$m_0 + m_1$:A'B'C'+A'B'C=A'B' $m_4 + m_6$:AB'C' +ABC' =AC' $m_7 + m_5$:ABC +AB'C =AC

Product Terms

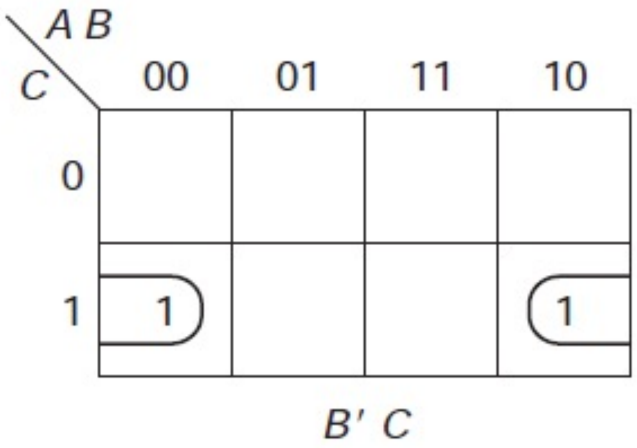
Also, outside columns (and outside rows when there are four rows) are adjacent. Thus,

– *m0 + m4: A'B'C' + AB'C' = B'C'*

– *m1 +m5:A'B'C+AB'C=B'C*



m0 +m4: A'B'C' + AB'C' = B'C'



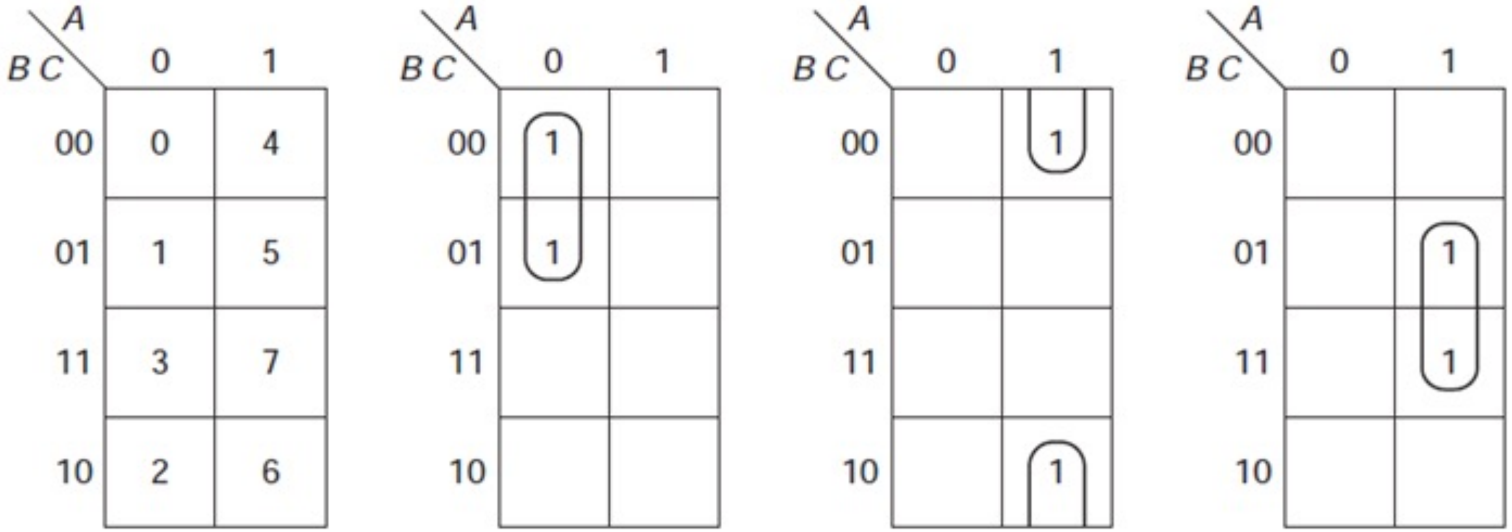
m1 +m5: A'B'C + AB'C = B'C

Vertical Oreintation

- Sometimes, more convenient to draw map in vertical orientation (that is, two columns and four rows) as shown in Map 3.6

- Both versions of the map produce the same results

Map 3.6 Vertical orientation of three-variable map.

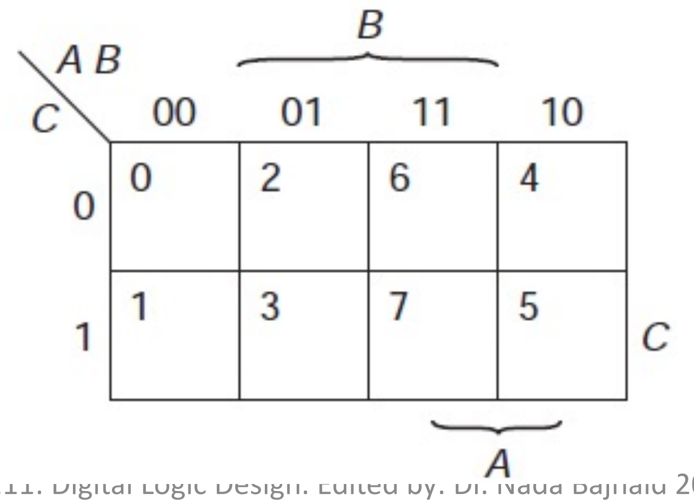


Map Columns Labeling

- In reading the map, it is useful to label the pairs of columns (in those arrangements where there are four columns) as shown in Map 3.7

- Thus, 1's in squares 4 and 6 are in the *A* columns and the *C* row (that is, not in the *C* row), producing the *AC* term as shown earlier

Map 3.7 Map with columns labeled.



Four Variables K-Map

- Four variable Karnaugh map contains **$2^4 = 16$** squares.
- A four-variable Karnaugh Map is shown below:

Map 3.8 The four-variable map.

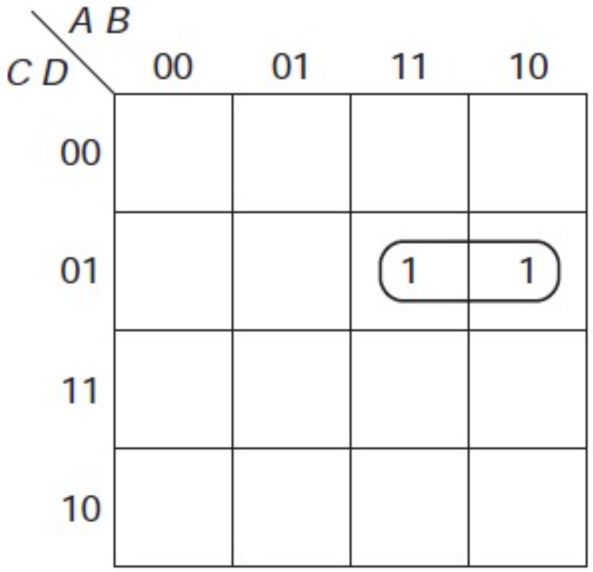
CD \ AB	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

CD \ AB	00	01	11	10
00	$A'B'C'D'$	$A'BC'D'$	$ABC'D'$	$AB'C'D'$
01	$A'B'C'D$	$A'BC'D$	$ABC'D$	$AB'C'D$
11	$A'B'CD$	$A'BCD$	$ABCD$	$AB'CD$
10	$A'B'CD'$	$A'BCD'$	$ABCD'$	$AB'CD'$

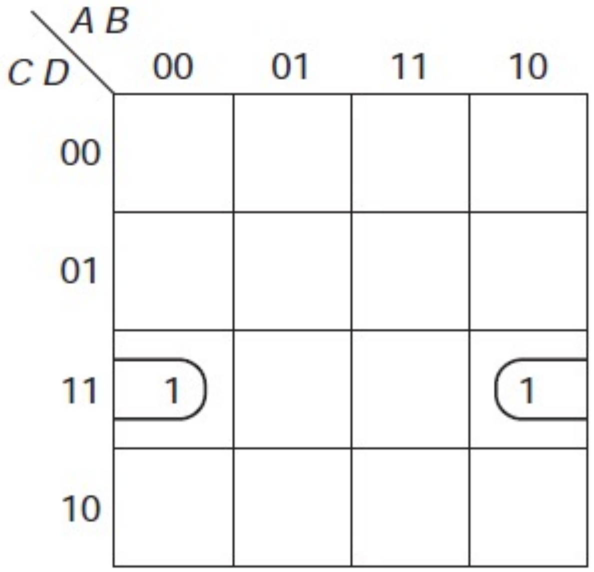
Four Variables K-Map

As with the three-variable map, 1's in two adjacent squares (where the top and bottom rows as well as the left and right columns are considered to be adjacent) correspond to a single product term (combined using P9a)

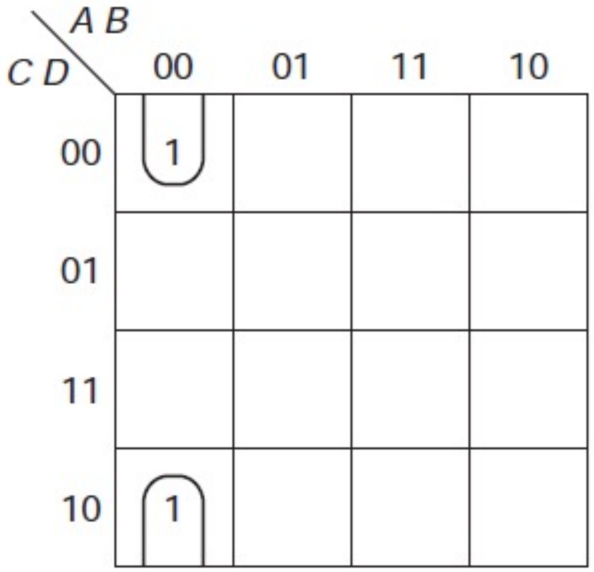
Example



A C′ D



B′ C D



A′ B′ D′

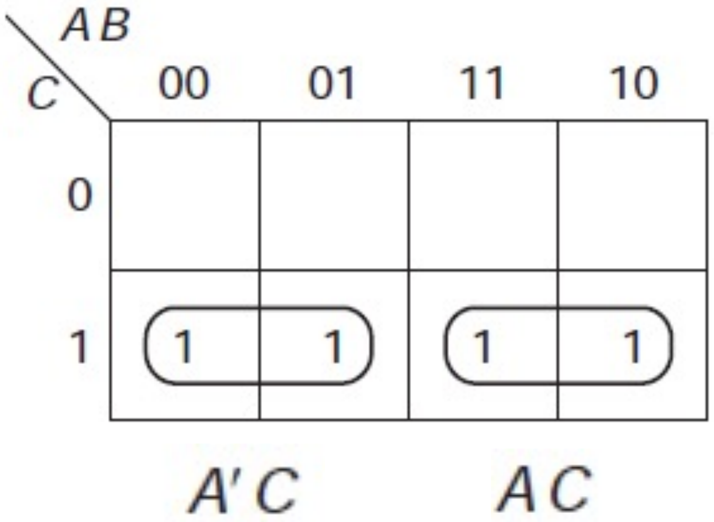
Group of Four 1s

- Up to this point, all of the product terms that have been shown correspond to 2 minterms combined using P9a
- These correspond to a product term with one literal missing, that is with
  - only two literals in a 3-variable function
  - three literals in a 4-variable function
- Let us next look at the maps of Map 3.9 with a group of four 1's

Group of Four 1s

On the map to the left, we have circled two groups of two, one forming the term *A'C* and the other forming the term *AC*

Map 3.9 A group of four 1's.



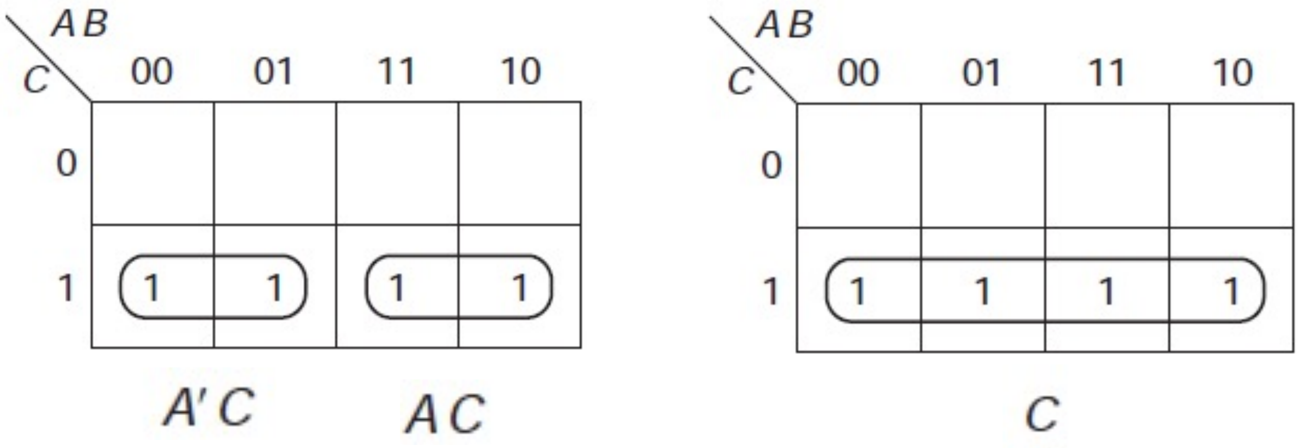
Group of Four 1s

- Obviously, P9a can be applied again to these two terms, producing

$$A'C + AC = C$$

- That is shown on the map to the right as a rectangle of four 1's

Map 3.9 A group of four 1's.



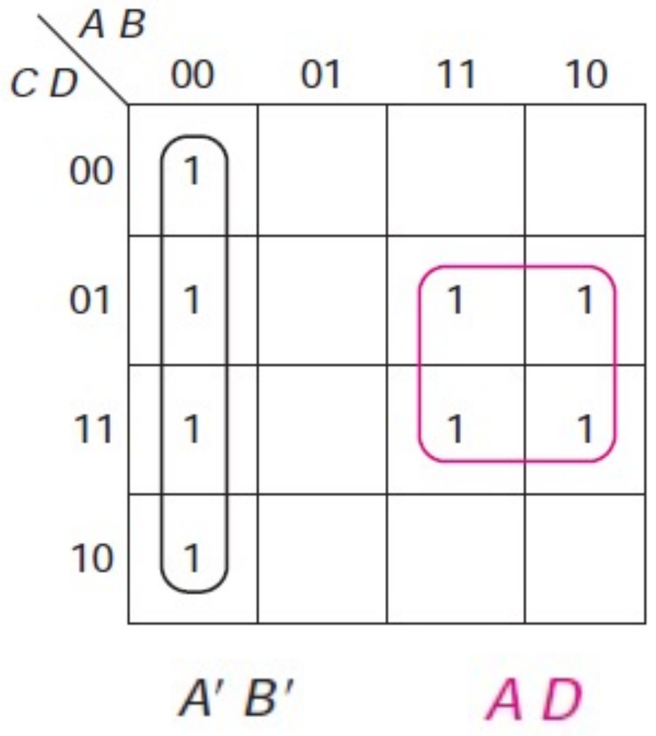
Group of Four 1s

In general, rectangles of four 1's will correspond to a product term with two of the variables missing, that is

– single literal for 3-variable problems
– two-literal term for 4-variable problems

Example

- Simplify the Boolean function: $F(A,B,C,D) = \sum(0, 1, 2, 3, 9, 11, 13, 15)$
- The K-Map for this function is as follows:

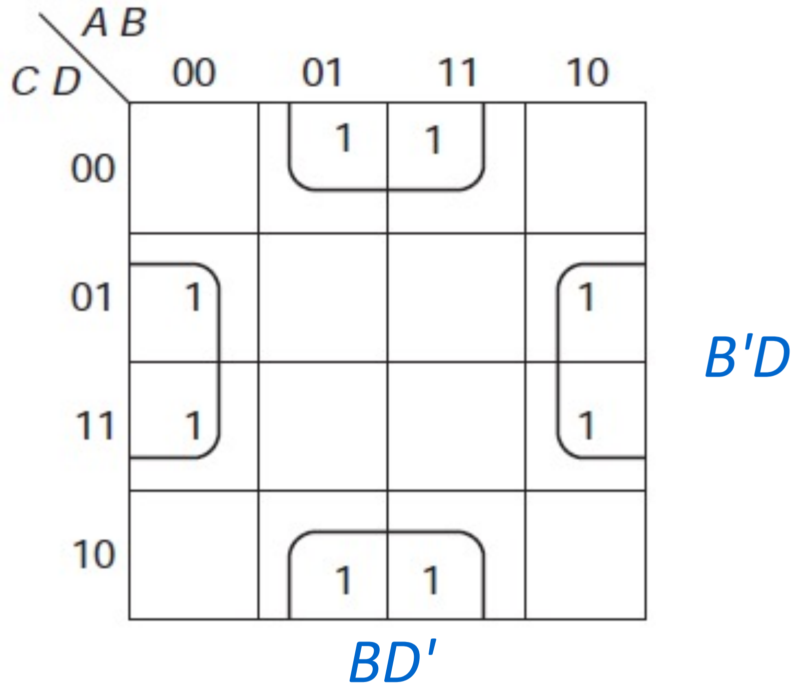


- So, the simplified function is $F = A'B' + AD$

Example

Simplify the Boolean function: $F(A,B,C,D) = \sum (1, 3, 4, 6, 9, 11, 12, 14)$

- The K-Map for this function is as follows:

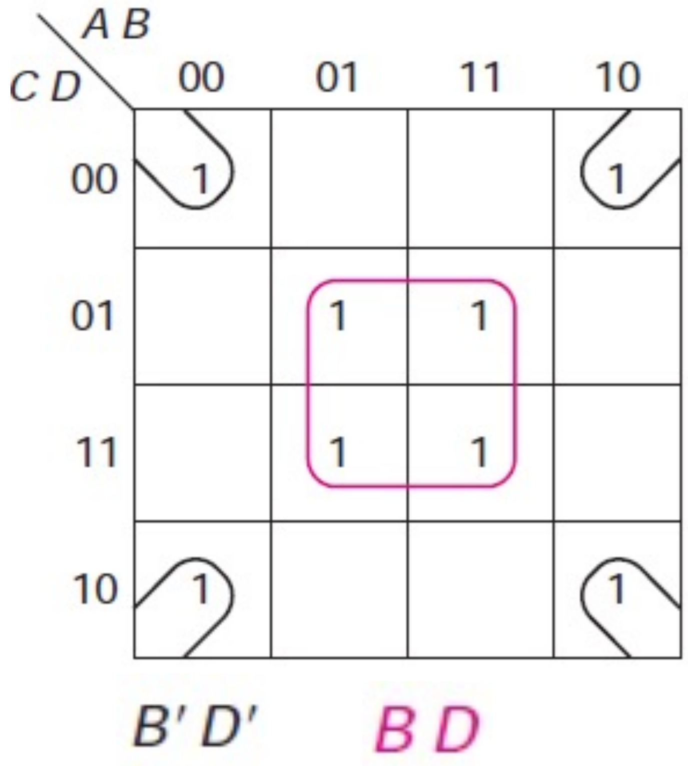


- So, the simplified function is $F = BD' + B'D$

Example

Simplify the Boolean function: $F(A,B,C,D) = \sum (0, 1, 2, 3, 9, 11, 13, 15)$

•The K-Map for this function is as follows:

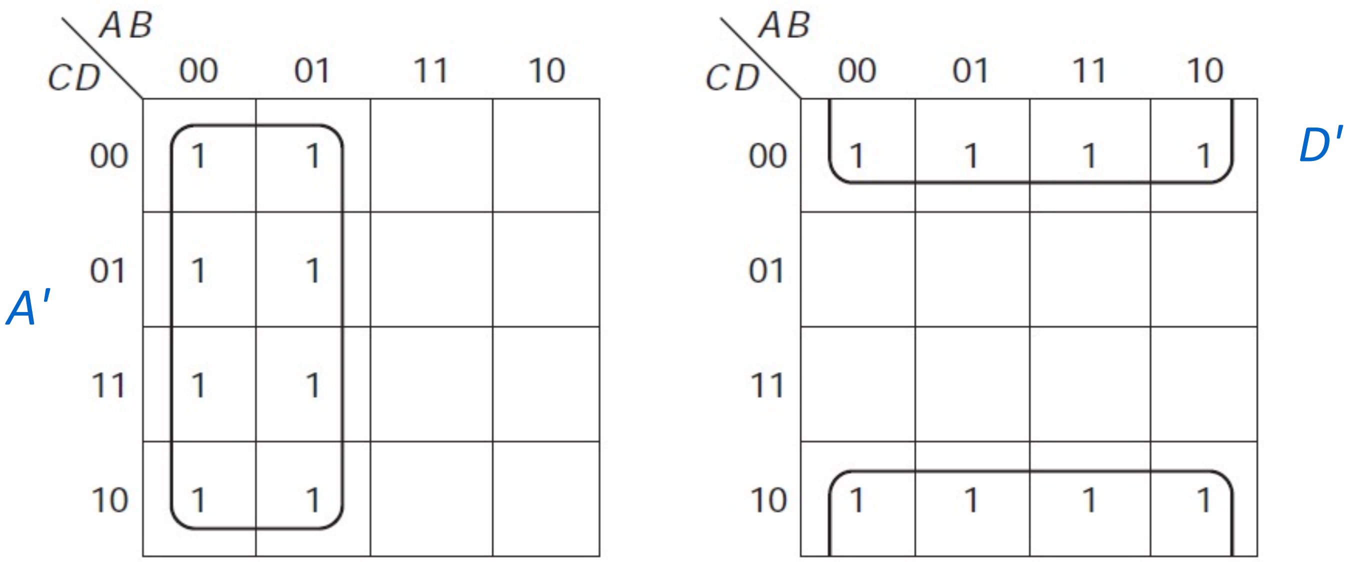


•So, the simplified function is $F = BD + B'D'$

Group of Eight 1s

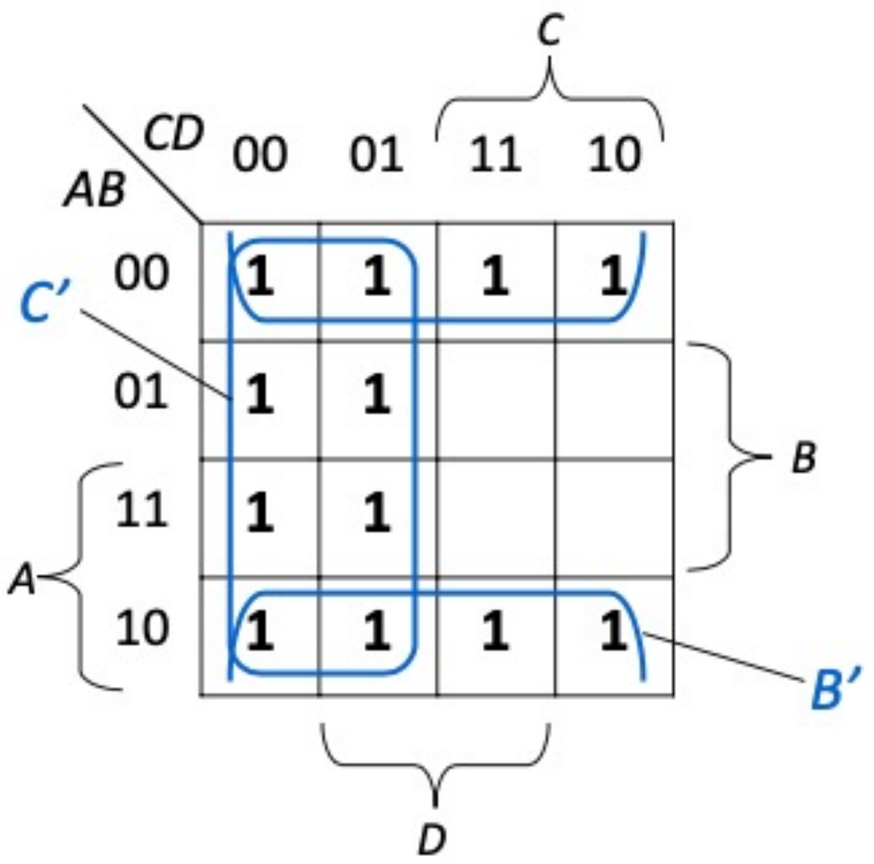
Two adjacent groups of four can be combined in a similar way to form a group of eight squares (with three of the literals missing)

Map 3.11 Groups of eight.



Example

- Simplify function: $F(A,B,C,D) = \sum (0, 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13)$
- K-Map for this function is as follows:



So, the simplified function is $F = B' + C'$

Example

Using K-Map, simplify the following function in Sum of Product as well as in Product of Sum

$$F(A,B,C,D) = A'B'+AC'D+B'CD'+AB'C'D$$

Solution: First complete the missing variables in each product term

$$\begin{aligned}
F &= A'B'(C + C') + AC'D(B + B') + B'CD'(A + A') + AB'C'D \\
&= A'B'C + A'B'C' + ABC'D + AB'C'D + AB'CD' + A'B'CD' + AB'C'D \\
&= A'B'C(D + D') + A'B'C'(D + D') + ABC'D + AB'C'D + AB'CD' + A'B'CD' + AB'C'D \\
&= A'B'CD + A'B'CD' + A'B'C'D + A'B'C'D' + ABC'D + AB'C'D + AB'CD' + A'B'CD' + AB'C'D \\
&= A'B'CD + A'B'CD' + A'B'C'D + A'B'C'D' + ABC'D + AB'C'D + AB'CD' + A'B'CD' + AB'C'D
\end{aligned}$$

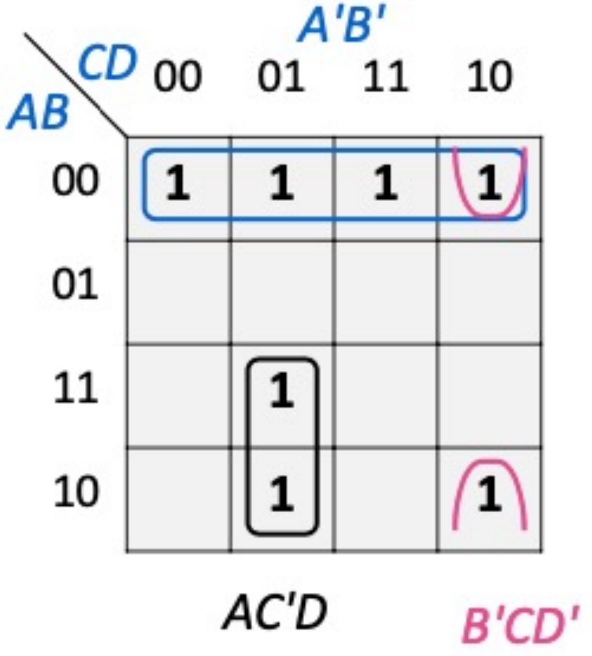
Example cont.

Now, replacing every primed (complemented) variable by 0 and uncomplemented variable by 1, we get

$$F = 0011 + 0010 + 0001 + 0000 + 1101 + 1001 + 1010 + 0010 + 1001$$

$$= m3 + m2 + m1 + m0 + m13 + m9 + m10 + m2 + m9$$

$$= \sum (0, 1, 2, 3, 9, 10, 13)$$



Its K-Map is as shown.

Combining 1's in map, we get

F = A'B' + B'CD' + AC'D which is the simplified expression in SOP.

Example cont.

To find the POS: fill in the remaining boxes by 0's and combine them to get

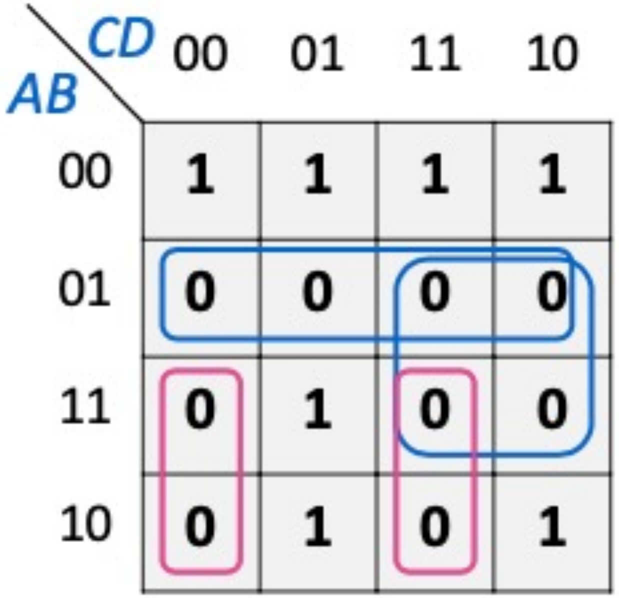
$$F' = A'B + BC + ACD + AC'D'$$

Taking complement on both sides

$$(F')' = (A'B + BC + ACD + AC'D')'$$

Applying DeMorgan's law, we get

$$F = (A + B')(B' + C')(A' + C' + D')(A' + C + D)$$



which is the required POS form.

Exercise

Simplify the following Boolean function

$$F(A, B, C, D) = \pi(0, 1, 4, 5, 8, 9, 14, 15)$$

in SOP as well as in POS forms.

Hint: Fill in the boxes of the given maxterms by 0's and the remaining boxes by 1's.

Mapping Function

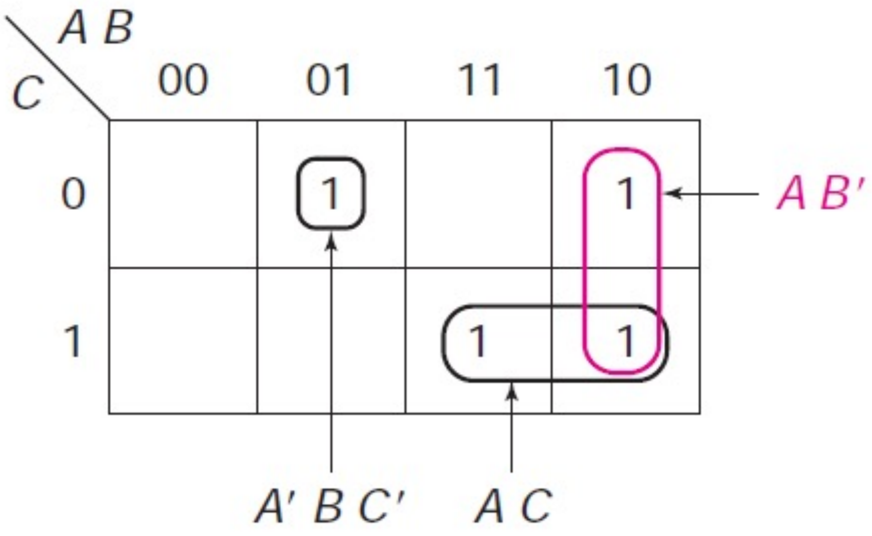
We can plot any function on the map

- either, we know the minterms, and use that form of the map
- or we put the function in SOP form and plot each of the product terms

Mapping Function

Map: $F = AB' + AC + A'BC'$

- F, with each of the product terms circled
- Each of two-literal terms corresponds to two squares on map (one of variables is missing)
  - AB' term is in 10 column
  - AC term is in C 1 row and in 11 and 10 columns (with common 1 in A position)



- Minterm $A'BC'$ corresponds to square, in 01 AB column and in 0 C row

Mapping Function

We could have obtained the same map by first expanding *F* to minterm form algebraically, that is,

$$
\begin{aligned}
F &= AB'(C' + C) + AC(B' + B) + A'BC' \\
&= AB'C' + AB'C + AB'C + ABC + A'BC' \\
&= m_4 + m_5 + m_5 + m_7 + m_2 \\
&= m_2 + m_4 + m_5 + m_7
\end{aligned}
$$

(removing duplicates and reordering)

C \ AB	00	01	11	10
0	0	2 1	6	4 1
1	1	3	7 1	5 1

Implicant

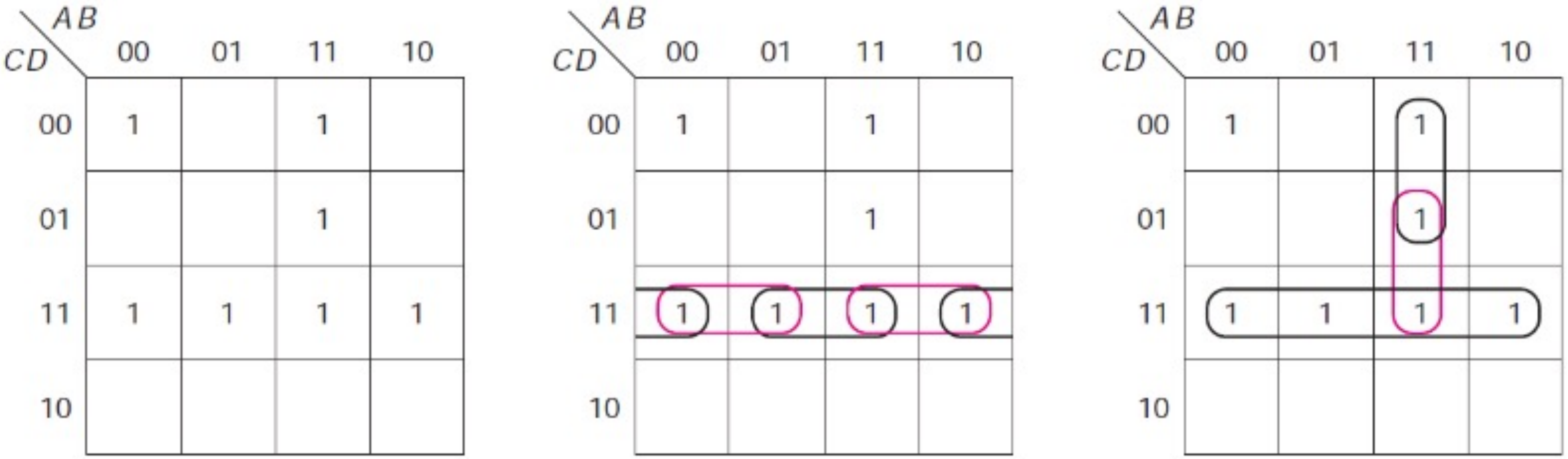
- An *implicant* of a function is a product term that can be used in an SOP expression for that function, that is, the function is 1 whenever implicant is 1
- From point of view of map, implicant is rectangle of 1, 2, 4, 8, . . . 1'; that rectangle may not include any 0's
- All minterms are implicants
- Example: For function $Y = AB + ABC + BC$
- – Implicants: AB, ABC, and BC because if any one of those terms are true, then Y is true

Implicant

Consider the function, *F,* of Map 3.12

- Second map shows the first four groups of 2
- Third map shows the other groups of 2 and the group of 4

Map 3.12 A function to illustrate definitions.



The implicants of F are

Minterms	*Groups of 2*	*Groups of 4*
$A'B'C'D'$		
$A'B'CD$		
$A'BCD$		
$ABC'D'$		
$ABC'D$		
$ABCD$		
$AB'CD$		

Implicant

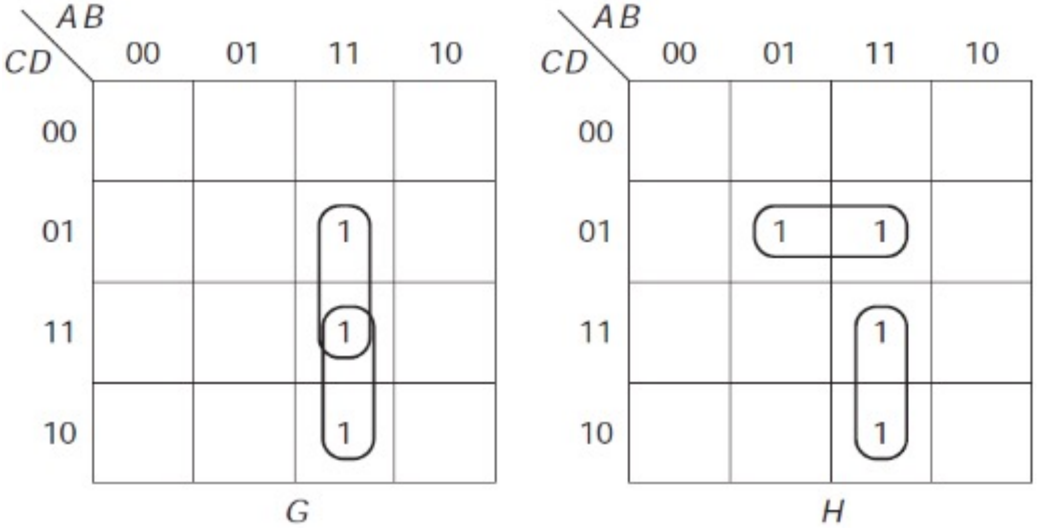
- Any SOP expression for *F* must be a sum of implicants
- Must choose enough implicants such that each of the 1’s of *F* are included in at least one of these implicants
- Such an SOP expression is sometimes referred to as a *cover* of *F*, and we sometimes say that implicant *covers* certain minterms

  e.g., ***ACD*** covers ***m*11** and ***m15***

- Implicants must be rectangular in shape and number of 1’s in rectangle must be a power of 2

Example

- *G:* three minterms, *ABC’D, ABCD, and ABCD’*
  - – Can be reduced no further than is shown on map, to *ABC* + *ABD*
  - - Group of two or four 1s not three



- *H:* same 3 minterms plus *A'BC'D*

– minimum expression is *BC'D* + *ABC*

– Note that *ABD* is also an implicant of *H*, but includes 1’s that are already included in the other terms

Prime Implicant

- A *prime implicant* is an implicant that (from point of view of map) is not fully contained in any one other implicant

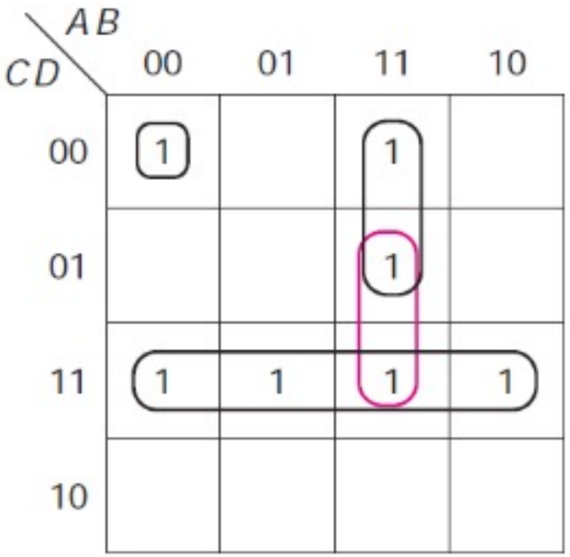
- e.g., it is a rectangle of two 1’s that is not part of a single rectangle of four 1’s

Prime Implicant

All prime implicants of *F* are circled: $A'B'C'D'$, ABC', ABD, and CD – Note that

- Only minterm that is not part of a larger group is m_0
- Other four implicants that are groups of two 1's are all part of group of four

Map 3.13 Prime implicants.



Prime Implicant

- From algebraic point of view, a prime implicant is an implicant such that if any literal is removed from that term, it is no longer an implicant
  - *ABCD* is prime implicant because *BCD*, *ACD*, *ABD*, and *ABC* are not implicants if we remove any literal from that term, we get a term that is 1 for some input combinations for function is to be 0
  - However, *ACD* is not a prime implicant since when we remove *A*, leaving *CD*, we still have an implicant

- Graphical approach to determine which implicants are prime implicants is easier than algebraic method of attempting to delete literals

Essential Prime Implicant

- An ***essential prime implicant*** is a prime implicant that includes at least one 1 that is not included in any other prime implicant

- If we were to circle all of the prime implicants of a function, the essential prime implicants are those that circle at least one 1 that no other prime implicant circles

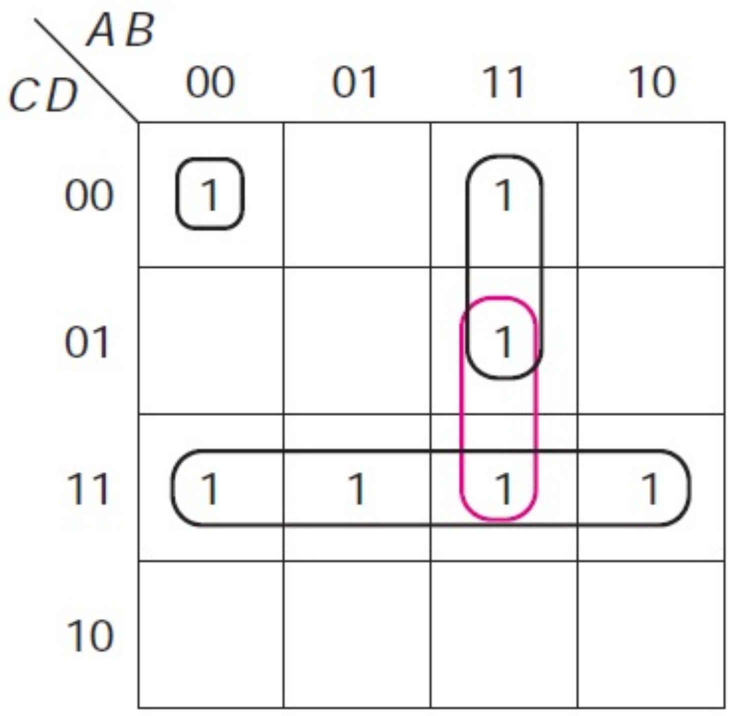
Essential Prime Implicant

- In the example, *A'B'C'D'*, *ABC'*, and *CD* are essential prime implicants; *ABD* is not

- Term *essential* is derived from idea that we must use that prime implicant in any minimum SOP expression

- There will often be prime implicant that is used in minimum solution that is not "essential."

Map 3.13 Prime implicants.



How to Find Essential Prime Implicant

- Single 1 on map represents prime implicant if it is not adjacent to any 1's
- 2 adjacent 1's form prime implicant if not within group of 4 adjacent squares
- 4 adjacent 1's form prime implicant if not within group of 8 adjacent squares
- Essential prime implicants are found by looking at each square marked with 1 and checking number of prime implicants that cover it
- Prime implicant is essential if it is the only prime implicant that covers minterm

Minimum SOP Expression Using K-Map

- In process of finding prime implicants, we will be considering each of the 1's on the map starting with the most *isolated 1's*

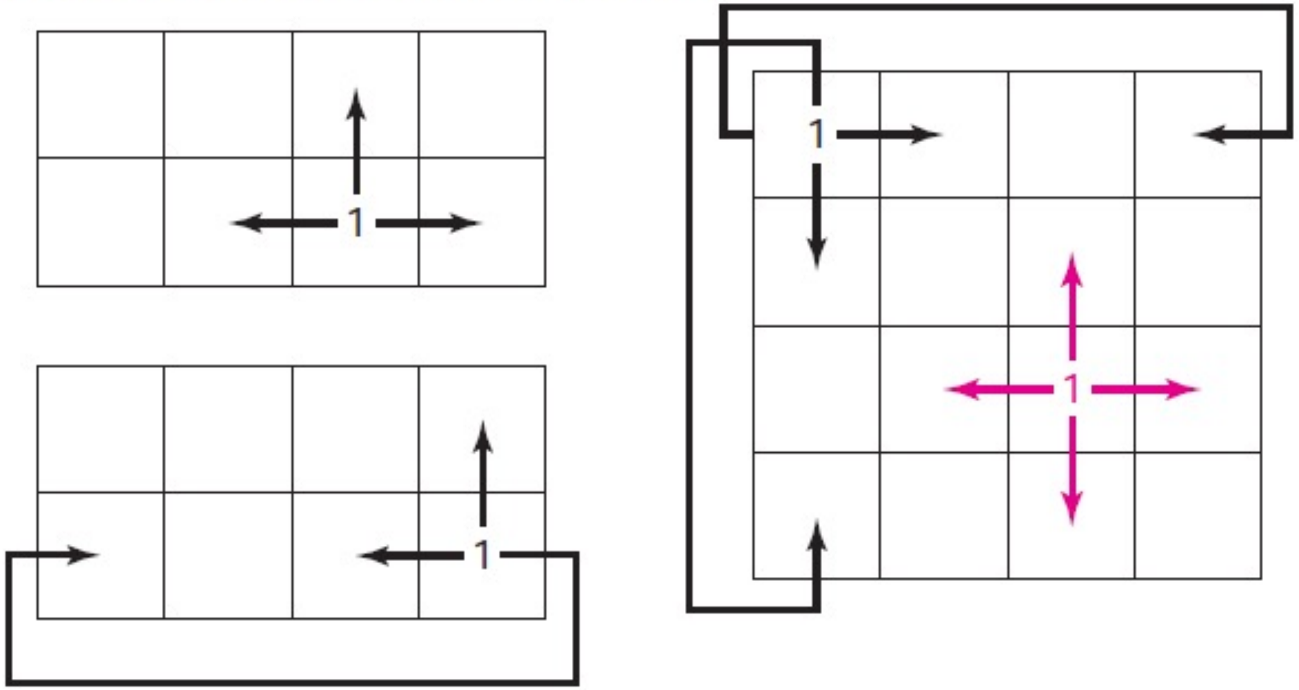
- Isolated means that there are few (or no) adjacent squares with a 1 in it

- In an *n-variable map, each square has n adjacent squares*

Adjacencies in K-Map

Examples for three- and four-variable maps are shown in Map 3.14.

Map 3.14 Adjacencies on three- and four-variable maps.



Map Method-1

1- Find all essential prime implicants

- Circle them on map and mark minterm(s) that make them essential with ★
  - Do this by examining each 1 on the map that has not already been circled
- Quickest to start with most isolated 1’s; fewest adjacent squares with 1’s

2- Find *enough* other prime implicants to cover function*; Do by* using 2 criteria

a. Choose a prime implicant that covers as many new 1’s (that is, those not already covered by a chosen prime implicant)

b. Avoid leaving isolated uncovered 1’s

Example

- m_0 has no adjacent 1's; it ($A'B'C'D'$) is a prime implicant
  - Indeed, it is an essential prime implicant, no other prime implicant covers this 1
- Next, m_{12} has only one adjacent 1
  - Those 1's are covered by prime implicant ABC'
  - No other prime implicant covers m_{12}, thus ABC' is essential
- Whenever we have a 1 with only one adjacent 1, that group of two is an essential prime implicant

CD \ AB	00	01	11	10
00	1		1	
01			1	
11	1	1	1	1
10				

Example cont.

- Each of 1's that have not yet been covered are part of the group of 4, *CD*
  - Each has 2 adjacent squares with 1's that are part of that group
  - Some squares, such as m_{15} may have more than two adjacent 1's
  - *CD* is essential because no other prime implicant covers m_3, m_7, or m_{11}

CD \ AB	00	01	11	10
00	1		1	
01			1	
11	1	1	1	1
10				

- Resulting in: $F = A'B'C'D' + ABC' + CD$

Another Example

Start looking at the most isolated 1, m_{11}

– It is covered only by the group of two shown, *wyz*

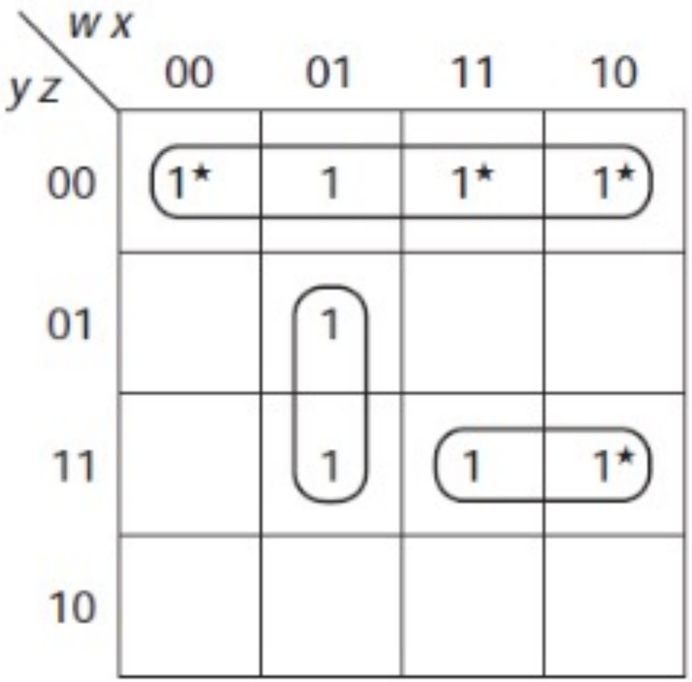
- The other essential prime implicant is *y'z'* because of m_0, m_8, or m_{12}

– None of these are covered by any other prime implicant each makes that prime implicant essential

yz \ wx	00	01	11	10
00	1	1	1	1
01		1		
11		1	1	1
10				

Example cont.

That leaves two 1’s uncovered; each of these can be covered by two different prime implicants, but the only way to cover them both with one term is shown on map below

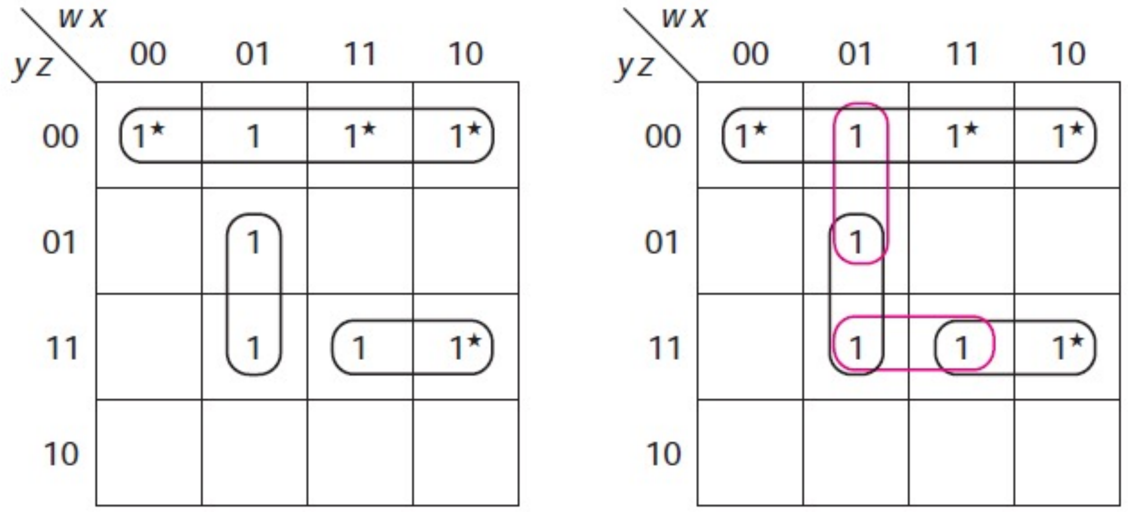


Thus, the minimum SOP solution is: $f = y'z' + wyz + w'xz$

Example cont.

- The other two prime implicants are *w'xy'* and *xyz*
  - they are redundant, since they cover no new 1's

- Even though *w'xz* must be used in a minimum solution, it does not meet the definition of an essential prime implicant; each of the 1's covered by it can be covered by other prime implicants



Example

$$F = x'yz' + x'yz\ xy'z' + xy'z + xyz$$

Two essential prime implicants

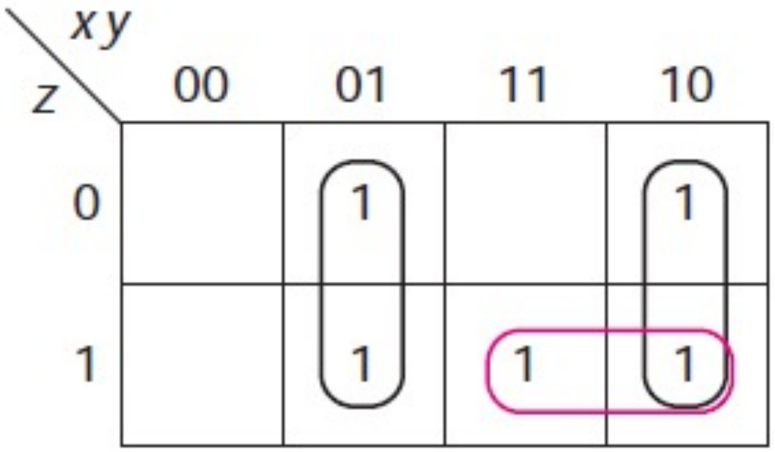
$z \backslash xy$	00	01	11	10
0		1		1
1		1	1	1

$z \backslash xy$	00	01	11	10
0		1*		1*
1		1	1	1

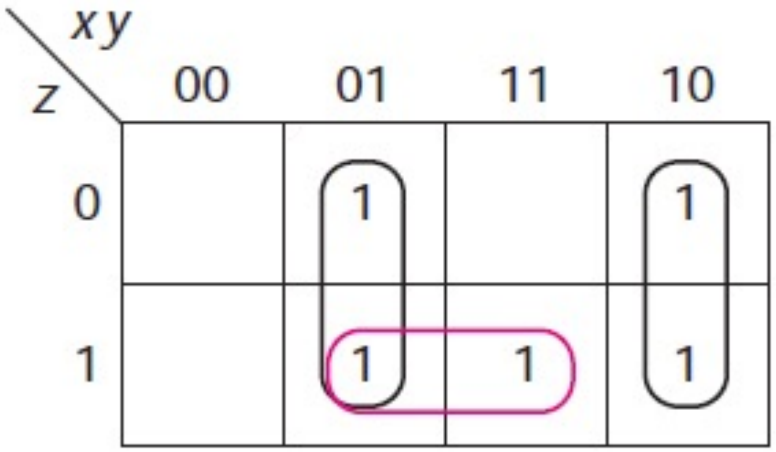
Example cont.

After finding the two essential prime implicants, m_7 is still uncovered.

The following maps show the two solutions

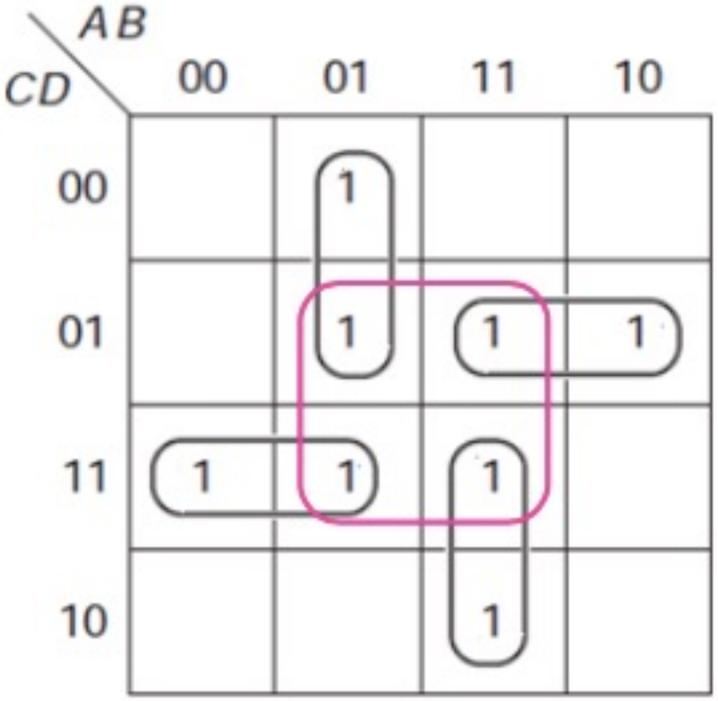


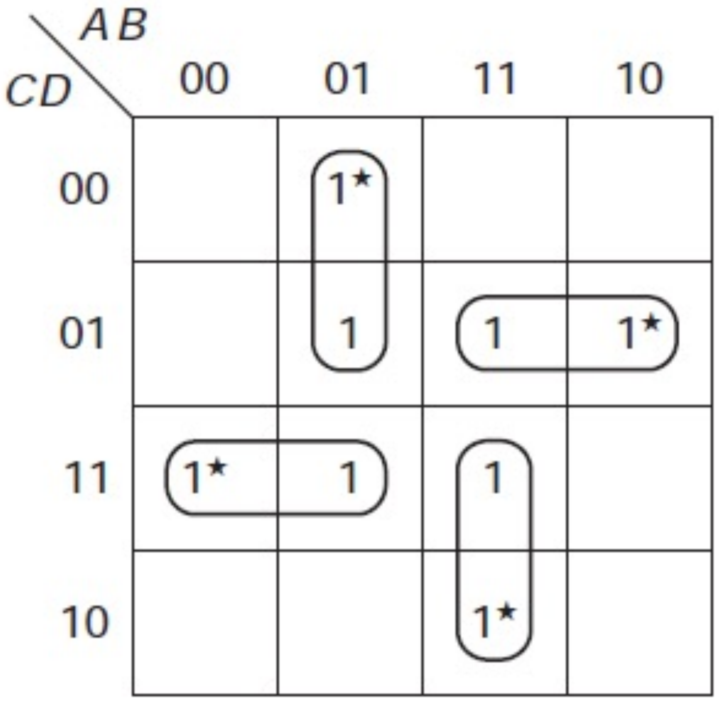
$x'y + xy' + xz$



$x'y + xy' + yz$

Example

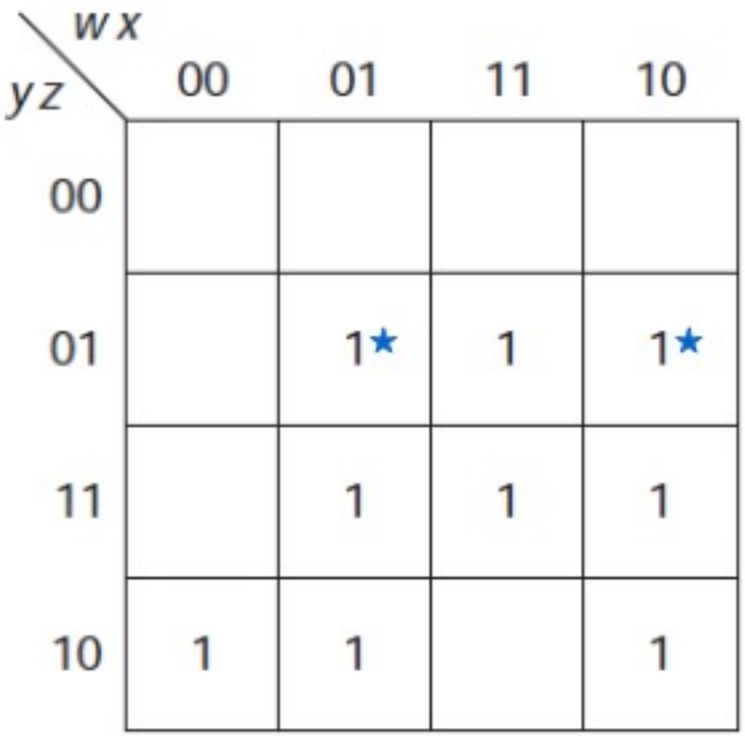


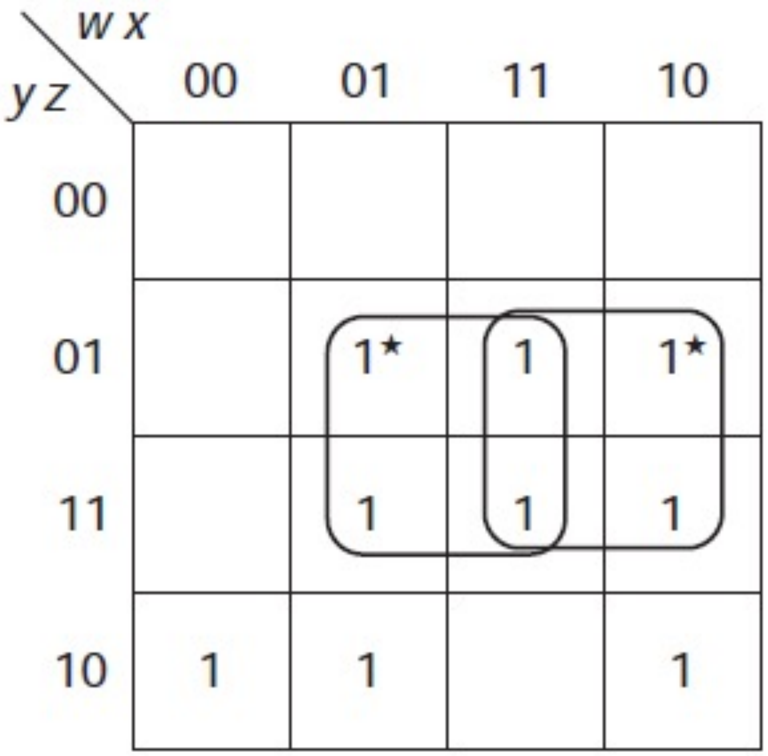


The minimum solution is $G=A'BC'+A'CD+ABC+AC'D$

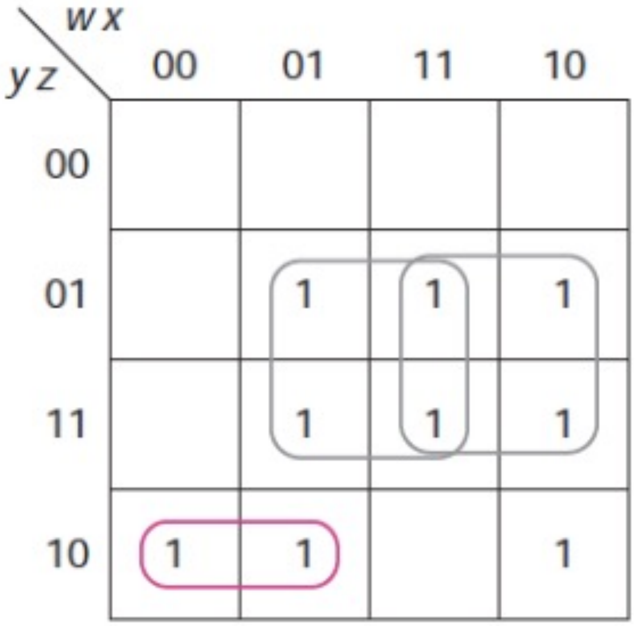
Example

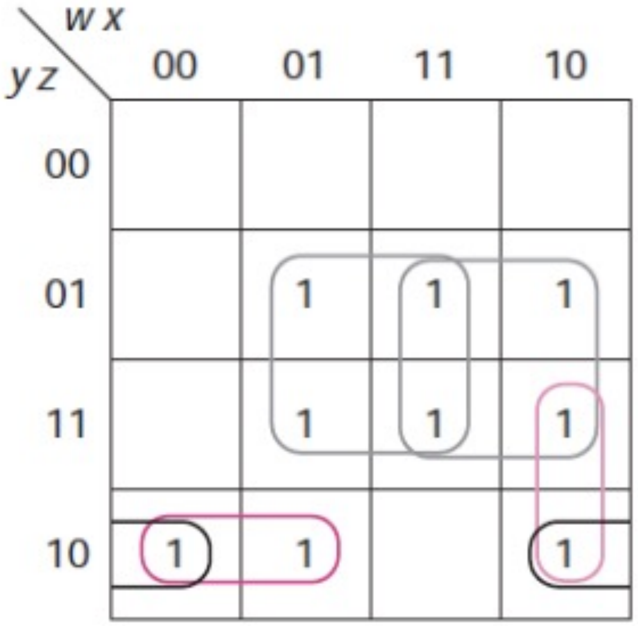
$$g(w,x,y,z)=\sum m(2,5,6,7,9,10,11,13,15)$$





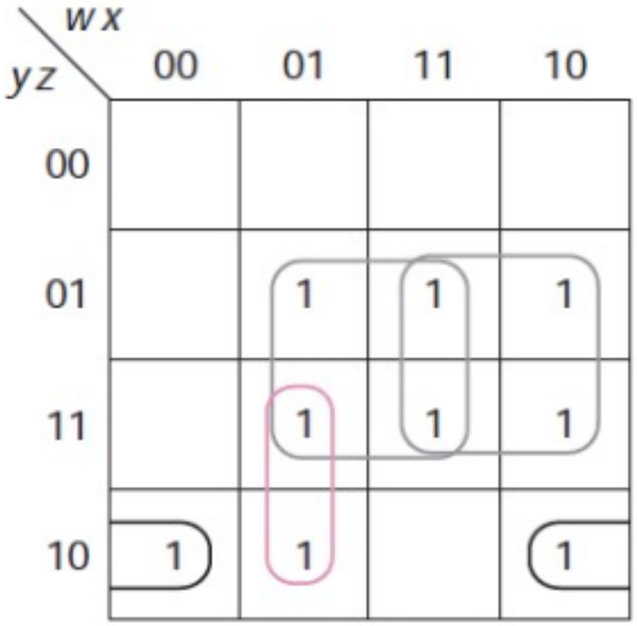
Example cont.





$g = xz + wz + w'yz' + wx'y$

$g = xz + wz + w'yz' + x'yz'$



$g = xz + wz + x'yz' + w'xy$

All three minimum solutions require four terms and 10 literals

Common Mistakes

- If there are multiple solutions, all minimum solutions must have the same number of terms and literals

- If, for example, you find a minimum solution with 3 terms and 7 literals
  - no solution with 4 terms is minimum, and
  - no solution with 3 terms and 8 literals is minimum

Map Method-2

1. Circle all of the prime implicants

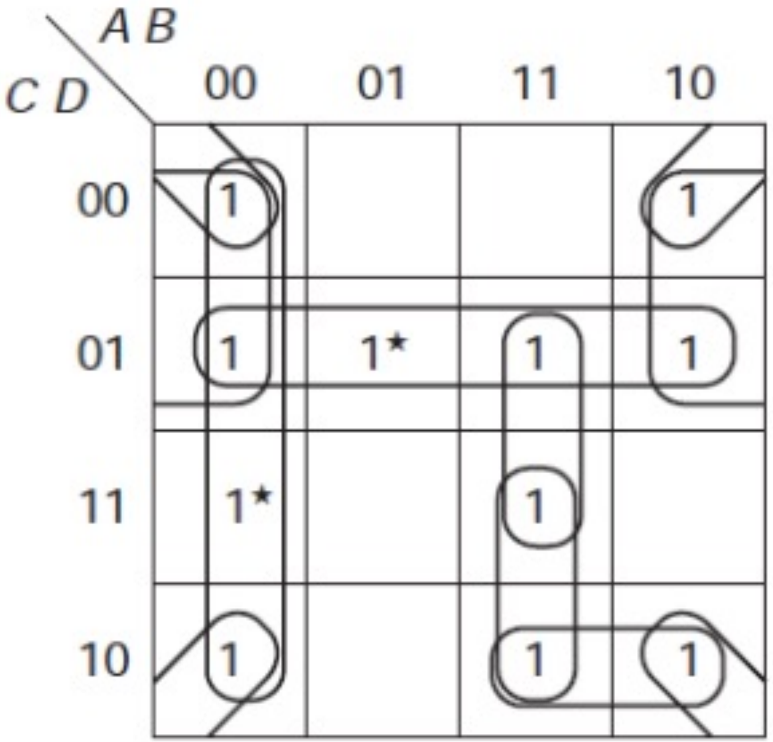
2. Select all essential prime implicants
   – Easily identified by finding 1’s that have only been circled once

3. Choose enough of the other prime implicants (as in Method 1)
   – These prime implicants have already been identified in step 1

Example

All of the prime implicants have been circled on the second map

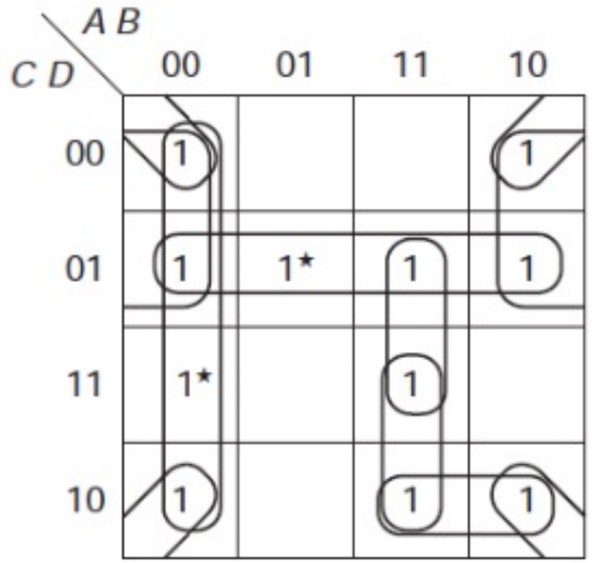
CD \ AB	00	01	11	10
00	1			1
01	1	1★	1	1
11	1★		1	
10	1		1	1

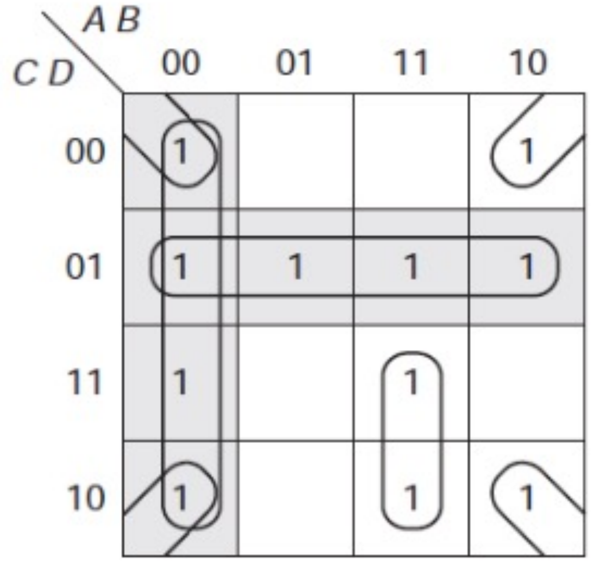


Example cont.

- m_0 circled 3 times
- m_3 and m_5 only circled once, thus, the prime implicants that cover them, *A'B'* and *C'D* are essential
- There are four 1's, each of which can be covered in two different ways, and five prime implicants not used yet
- No prime implicant covers more than two new 1's; thus, we need at least two more terms
- Of the groups of four, only *B'D'* covers two new 1's; *B'C'* covers only one
- Having chosen the first group, we must use *ABC* to cover the rest of the function, producing

$$F=A'B'+ C'D+B'D'+ABC$$





Don’t Cares

- Finding minimum solutions for functions with don’t cares does not significantly change the methods we developed previously

- We need to modify slightly the definitions of an implicant and a prime implicant, and clarify the definition of an essential prime implicant

Don’t Cares

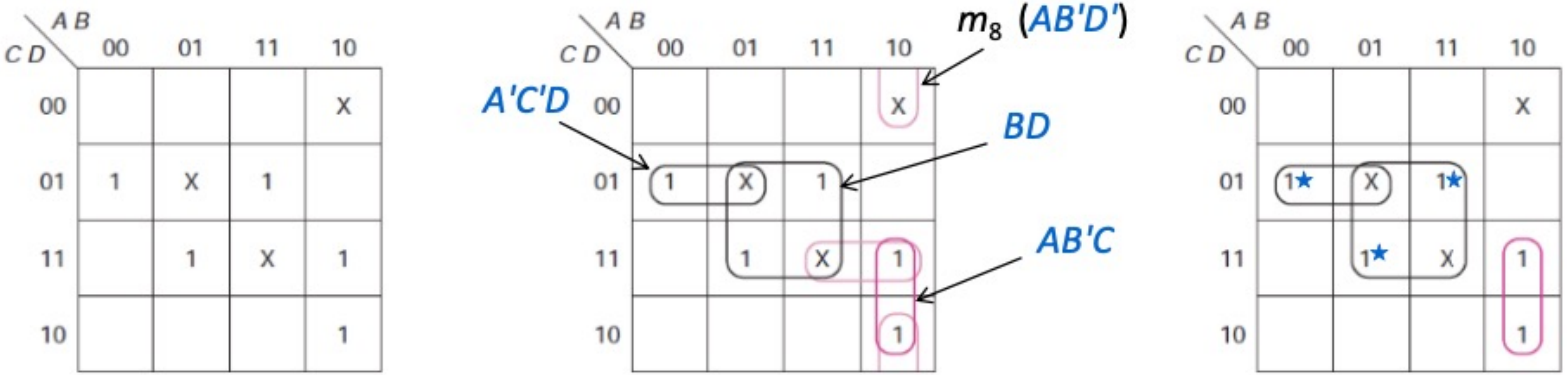
- ***Implicant*** is a rectangle of 1, 2, 4, 8, . . . 1’s or X’s (containing no 0’s)
- ***Prime implicant*** is an implicant not included in any one larger rectangle
  - From point of view of finding prime implicants, X’s are treated as 1’s
- ***Essential prime implicant*** is a prime implicant that covers at least one 1 not covered by any other prime implicant (as always)
  - Don’t cares (X’s) do not make a prime implicant essential

Example

$F(A,B,C,D)= \sum m(1,7,10,11,13) + \sum d(5,8,15)$

– If all of X were made 1's, we would need a fourth term to cover m_8, making

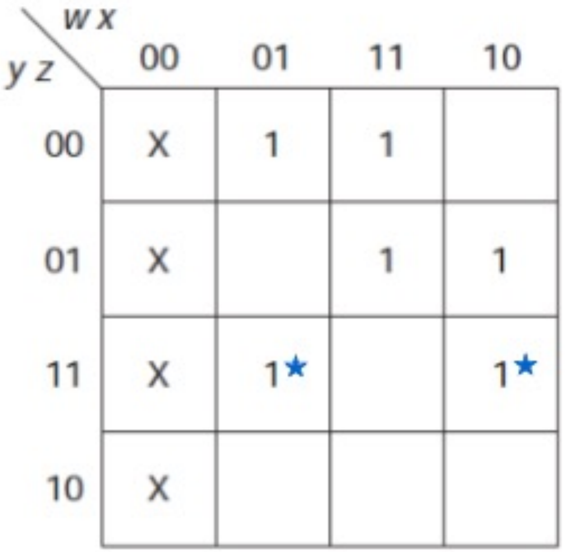
- $F = BD + A'C'D + AB'C + AB'D'$ or
- $F = BD + A'C'D + ACD + AB'D'$

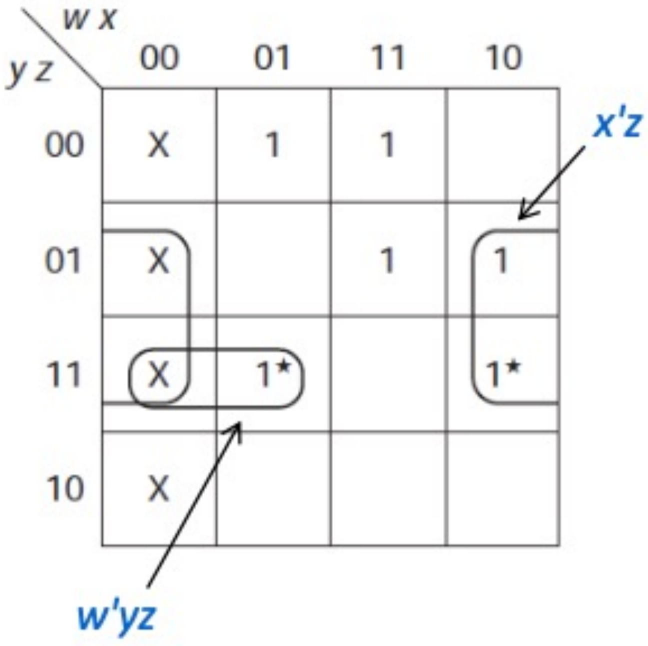


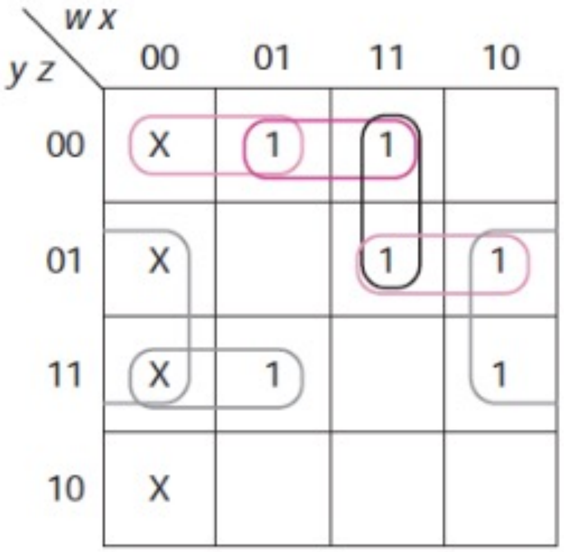
Minimum solution: $F = BD + A'C'D + AB'C$

Example

$$F(w,x,y,z)=\sum m\,(4,7,9,11,12,13)+\sum d\,(0,1,2,3)$$

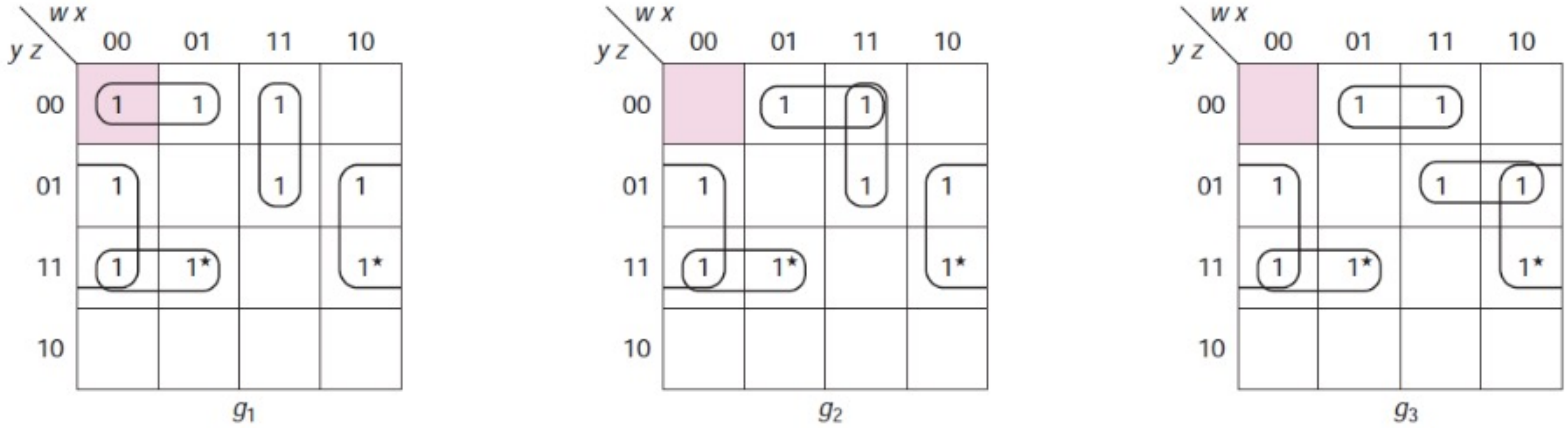






Example cont.

Minimum solution: Each using 4 terms and 11 literals



$g_1 = x'z + w'yz + w'y'z' + wxy'$ $\quad g_2 = x'z + w'yz + xy'z' + wxy'$ $\quad g_3 = x'z + w'yz + xy'z' + wy'z$

Example cont.

Notices

– Group of 4 X's, *w'x'*, *is a prime implicant*

- Rectangle of four 1's or X's, but it is not essential

  – It does not cover any 1's not covered by some other prime implicant

$yz \backslash wx$	00	01	11	10
00	X	1	1	
01	X		1	1
11	X	1		1
10	X			

– Surely, a prime implicant made up of all don't cares would never be used, since that would add a term to the sum without covering any additional 1's

Example

Suppose a Boolean function is given with minterms and don't care conditions as $F(A, B, C, D) = \sum m\,(0, 2, 3, 4, 8, 10, 12, 13, 14, 15) + \sum d\,(7, 9)$

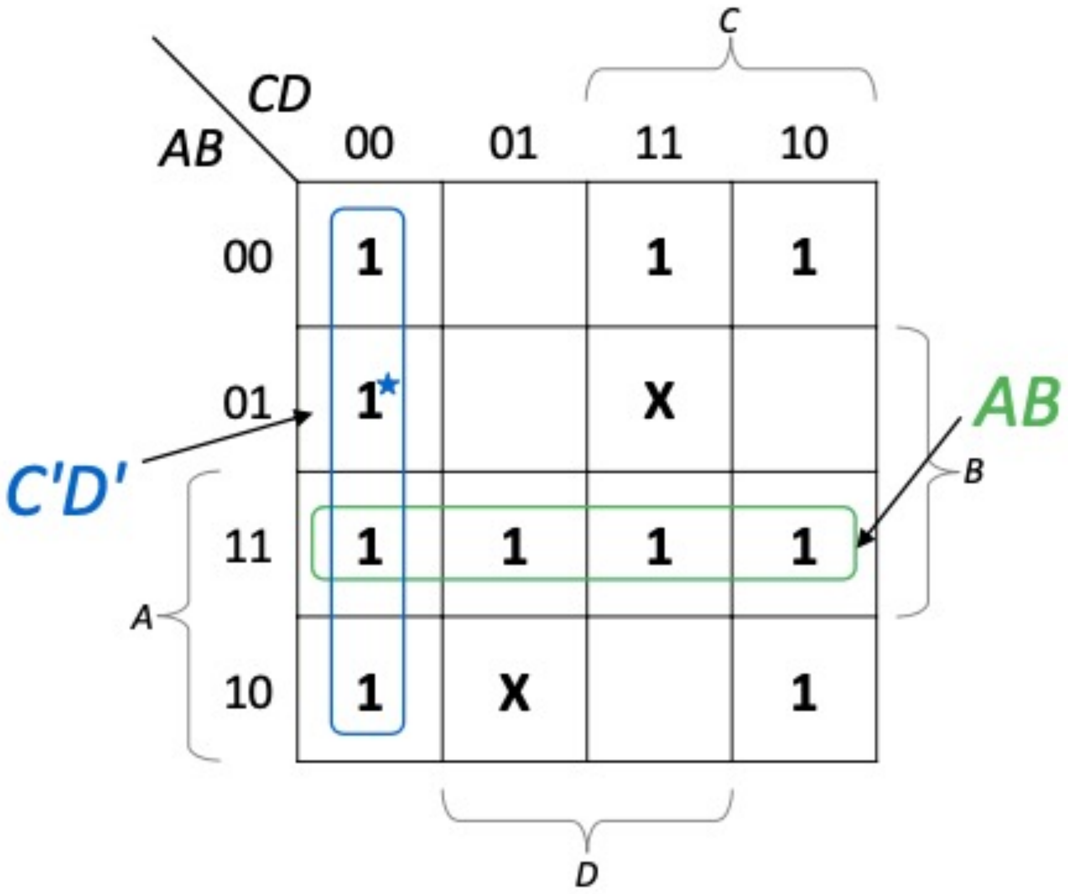
- K-Map is shown

AB \ CD	00	01	11	10
00	1		1	1
01	1		X	
11	1	1	1	1
10	1	X		1

Example cont.

Essential prime implicants is: *C'D'*

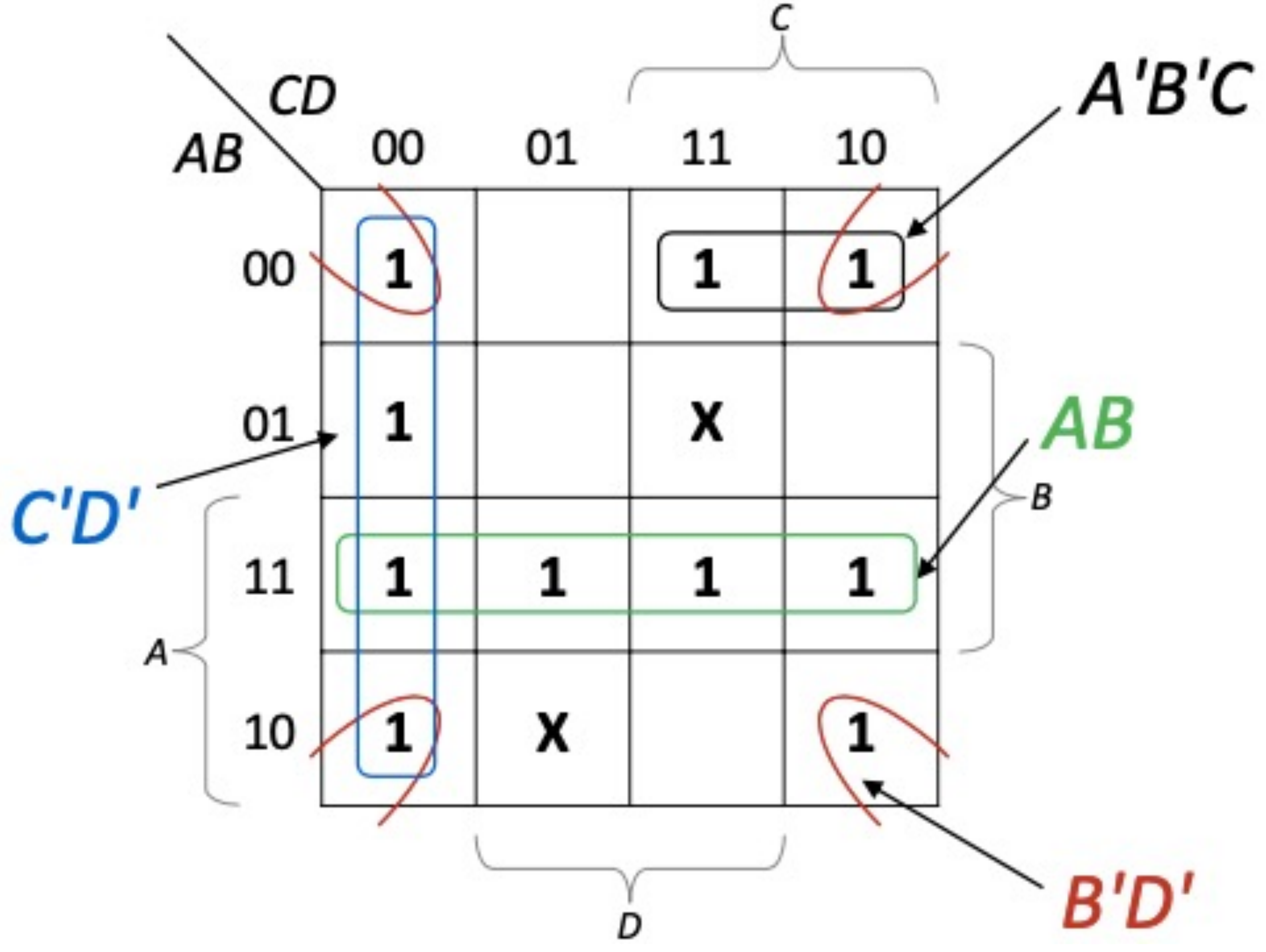
F =AB+C'D'+



Example cont.

Thus, we can have the following three solutions:

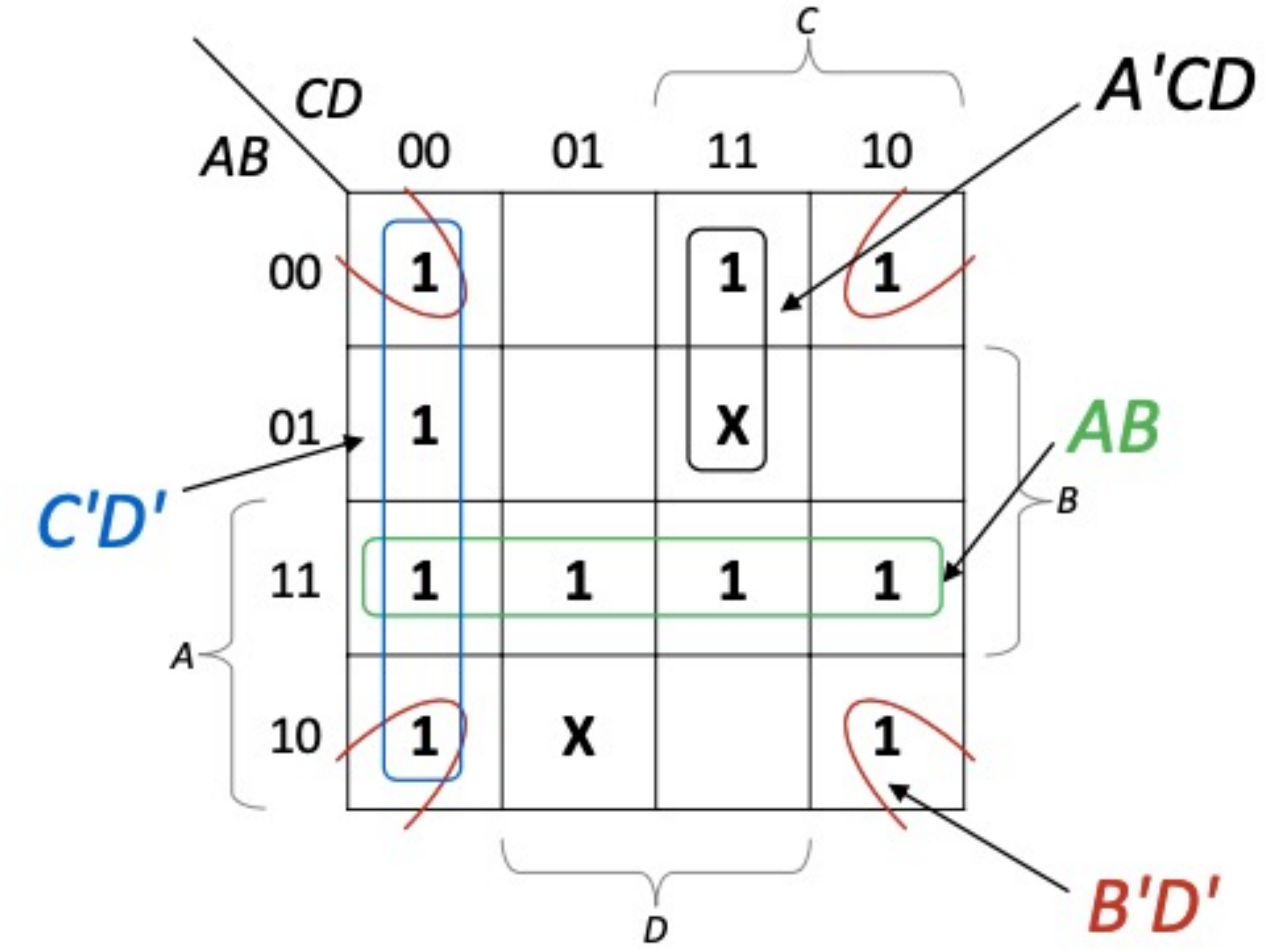
$F1 = AB + C'D' + B'D' + A'B'C$



Example cont.

Thus, we can have the following three solutions:

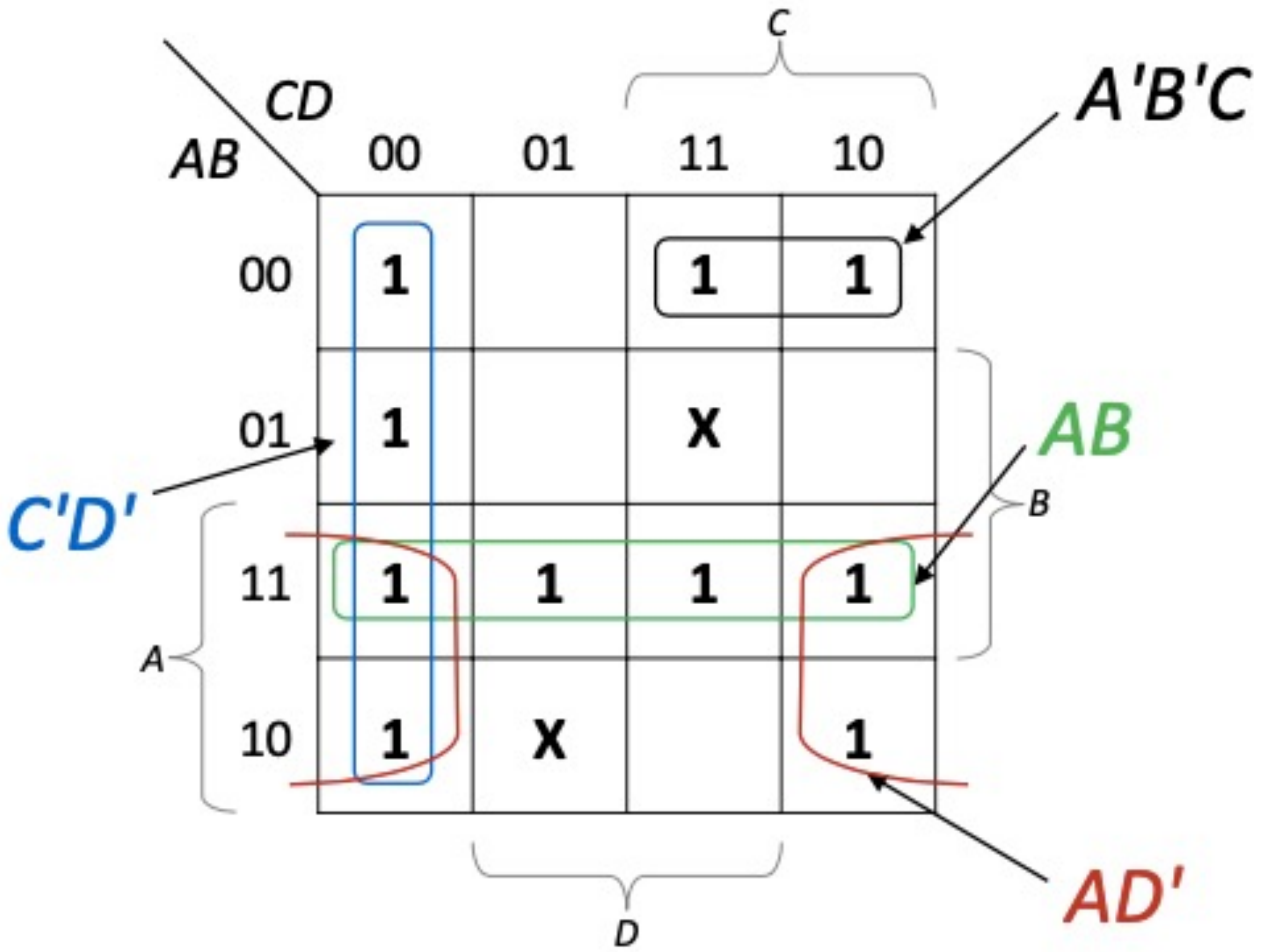
$F_2 = AB + C'D' + B'D' + A'CD$



Example cont.

Thus, we can have the following three solutions:

$F_3 = AB + C'D' + AD' + A'B'C$

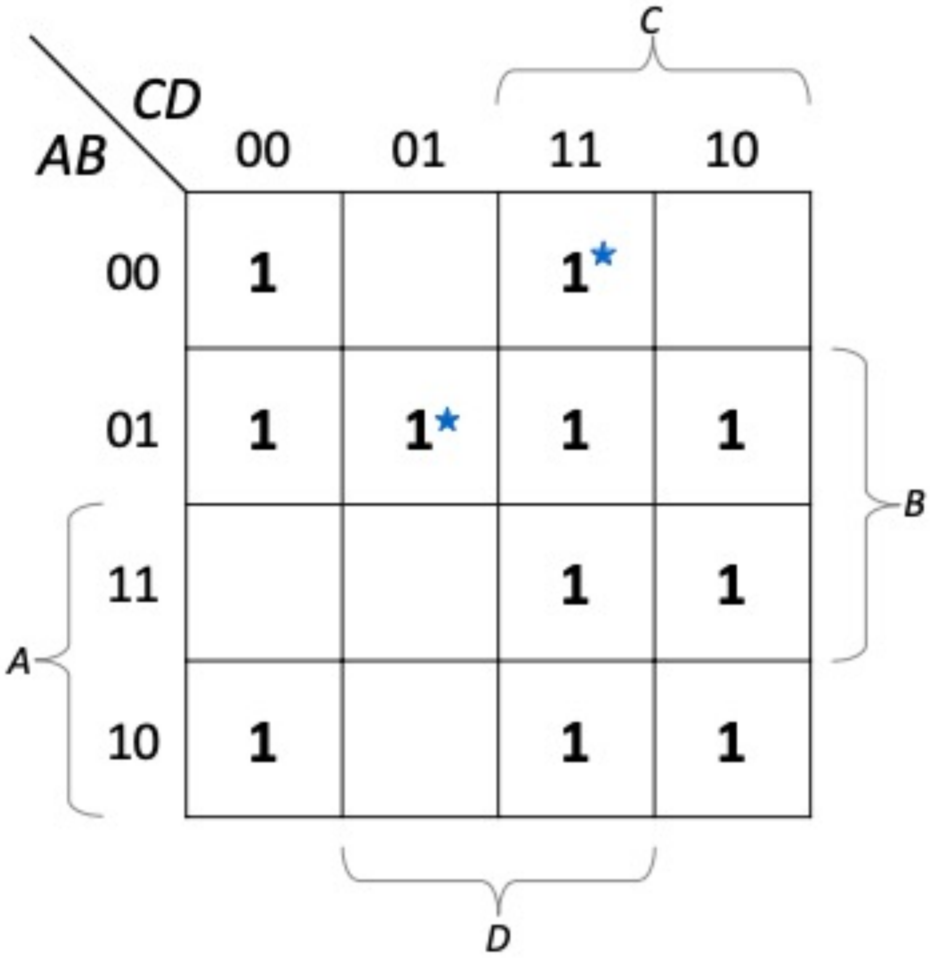


Another Method to Get Simplified Function

- Don’t cares give us another option for solving map problems for functions with or without don’t cares
- At any point in the process of using either Map Method 1 or 2, we can replace all 1’s covered (by the terms already chosen) by don’t cares
- That highlights the 1’s remaining to be covered
- We then need to choose enough terms to cover the remaining 1’s
- This works because the 1’s already covered can be used again (as part of a term covering some new 1’s), but need not be

Example

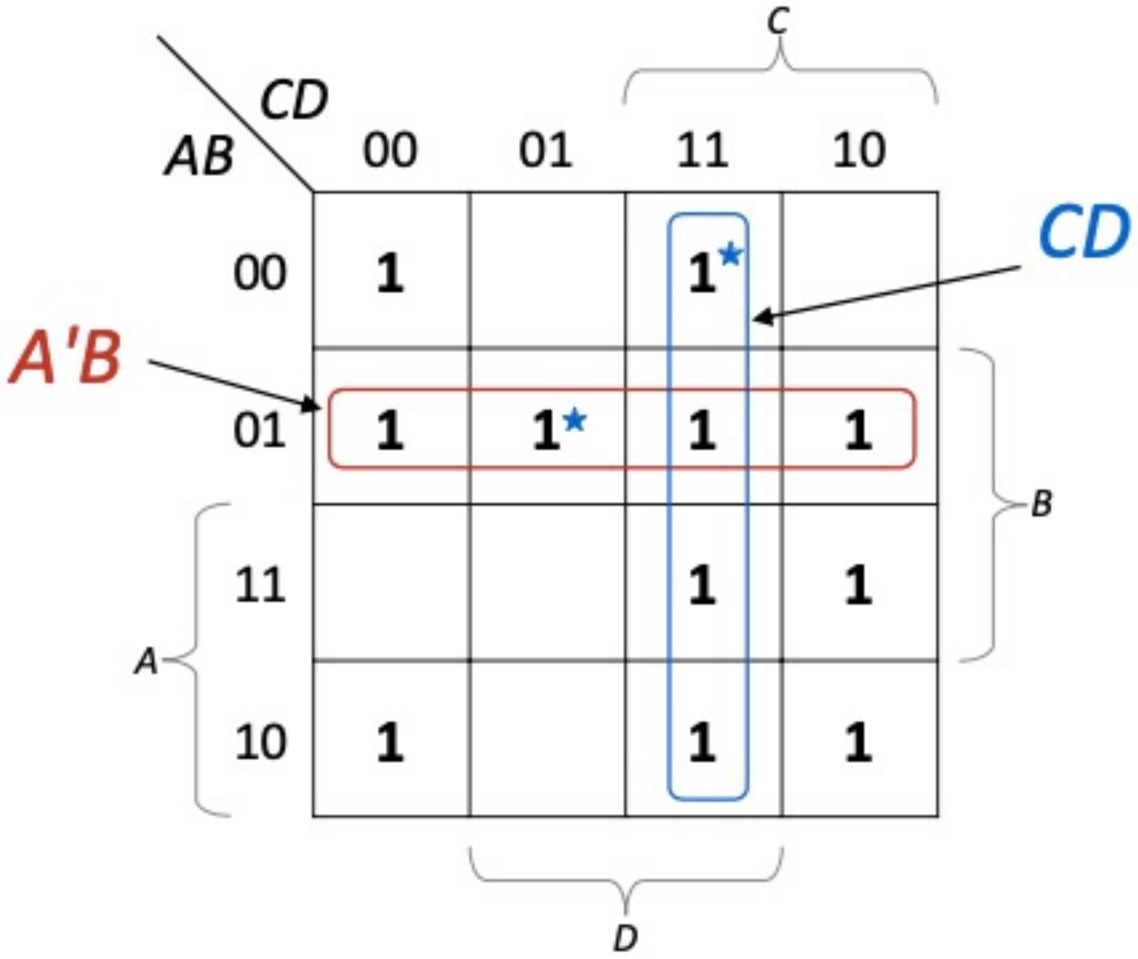
$F(A, B, C, D) = \sum m\,(0, 3, 4, 5, 6, 7, 8, 10, 11, 14, 15)$



The simplified function is: $F =$

Example cont.

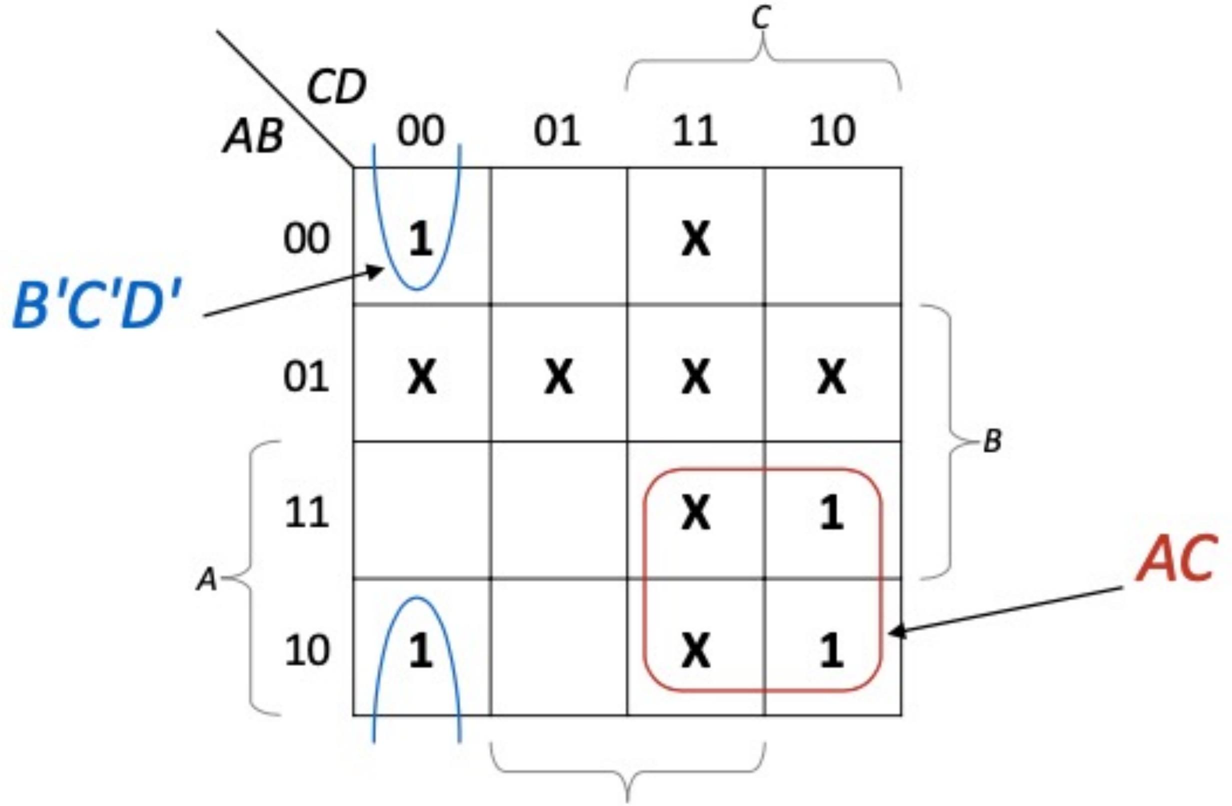
$F(A, B, C, D) = \sum m\ (0, 3, 4, 5, 6, 7, 8, 10, 11, 14, 15)$



The simplified function is: $F = A'B + CD$

Example cont.

$F(A, B, C, D) = \sum m\ (0, 3, 4, 5, 6, 7, 8, 10, 11, 14, 15)$



The simplified function is: $F = A'B + CD + AC + B'C'D'$

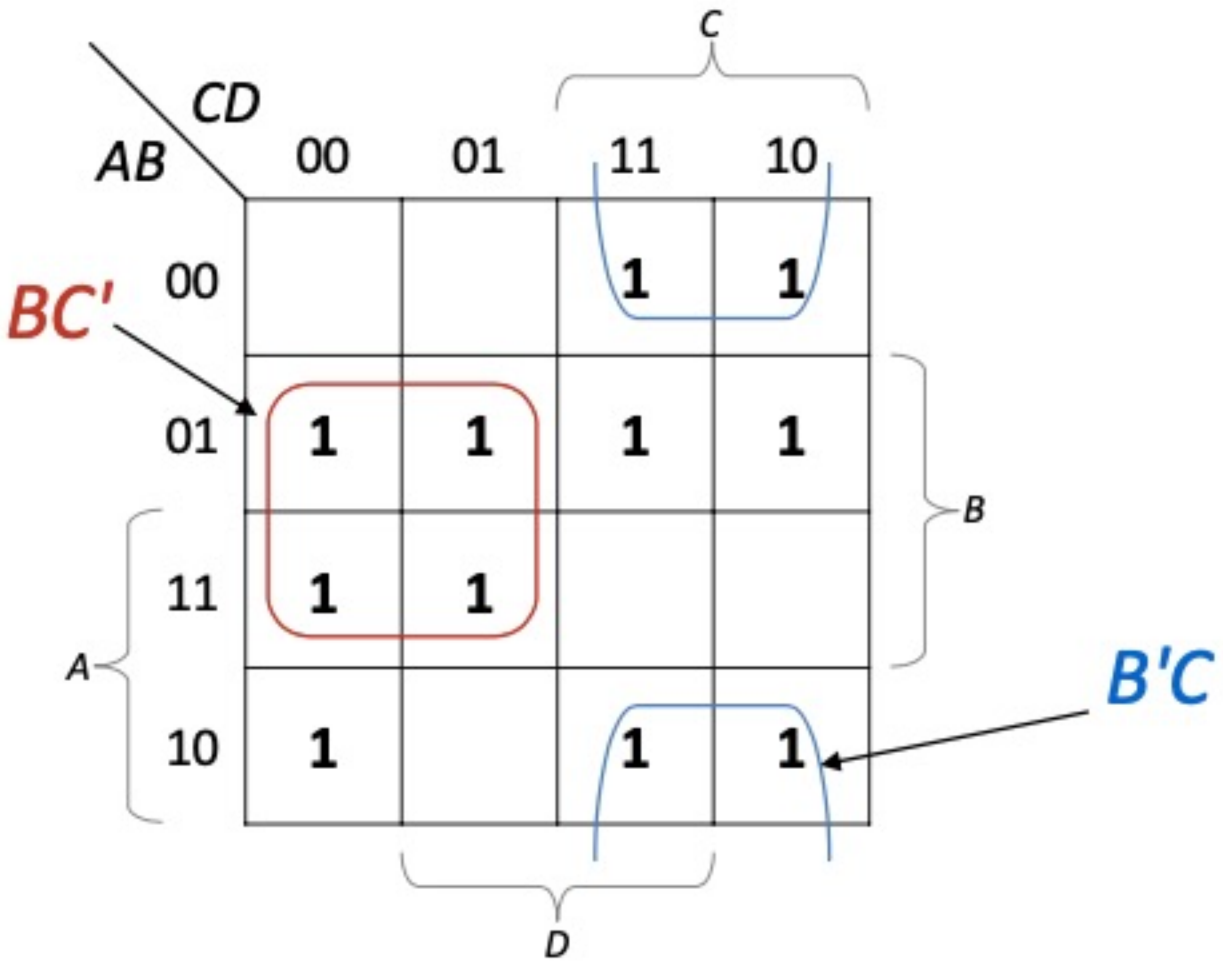
Another Example

$F(A, B, C, D) = \sum m\,(2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13)$

AB \ CD	00	01	11	10
00			1	1
01	1	1	1	1
11	1	1		
10	1		1	1

Example cont.

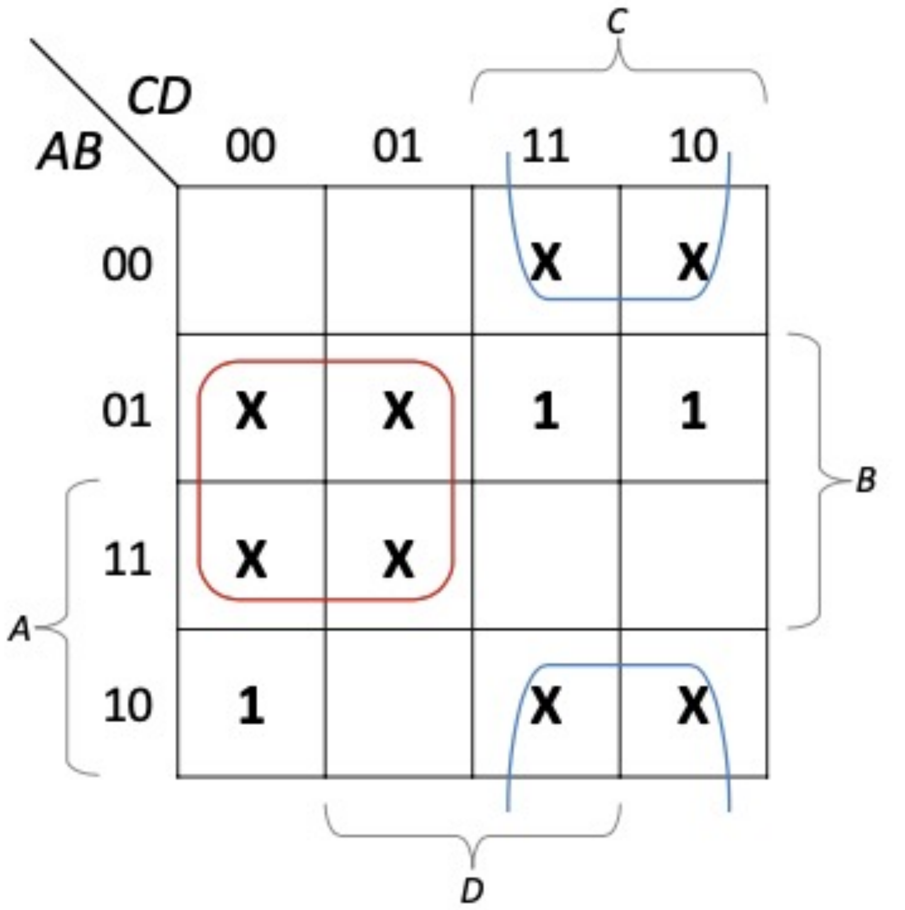
Essential prime implicants:
BC' and *B'C*



Example cont.

Essential prime implicants: *BC'* and *B'C*

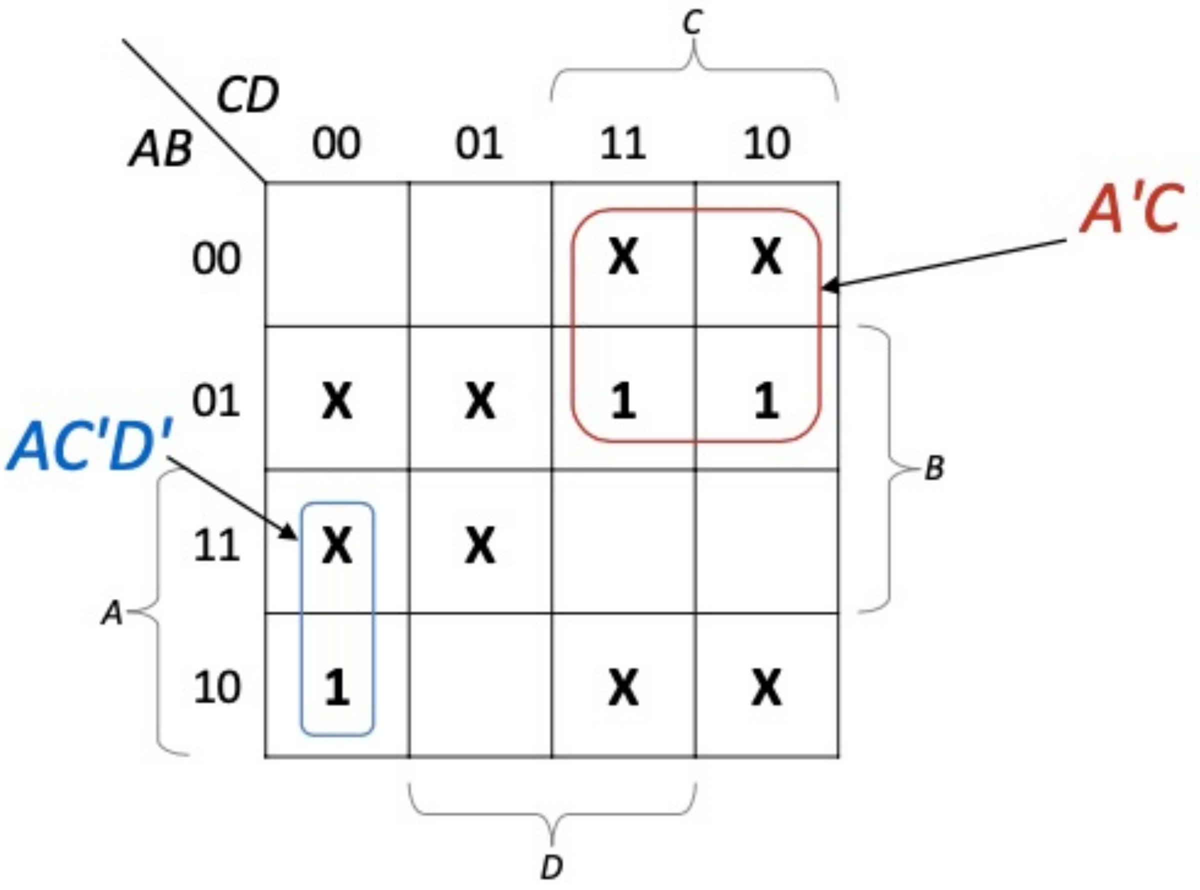
– 1's covered by them are changed to don't cares



Example cont.

The simplified function has four solutions:

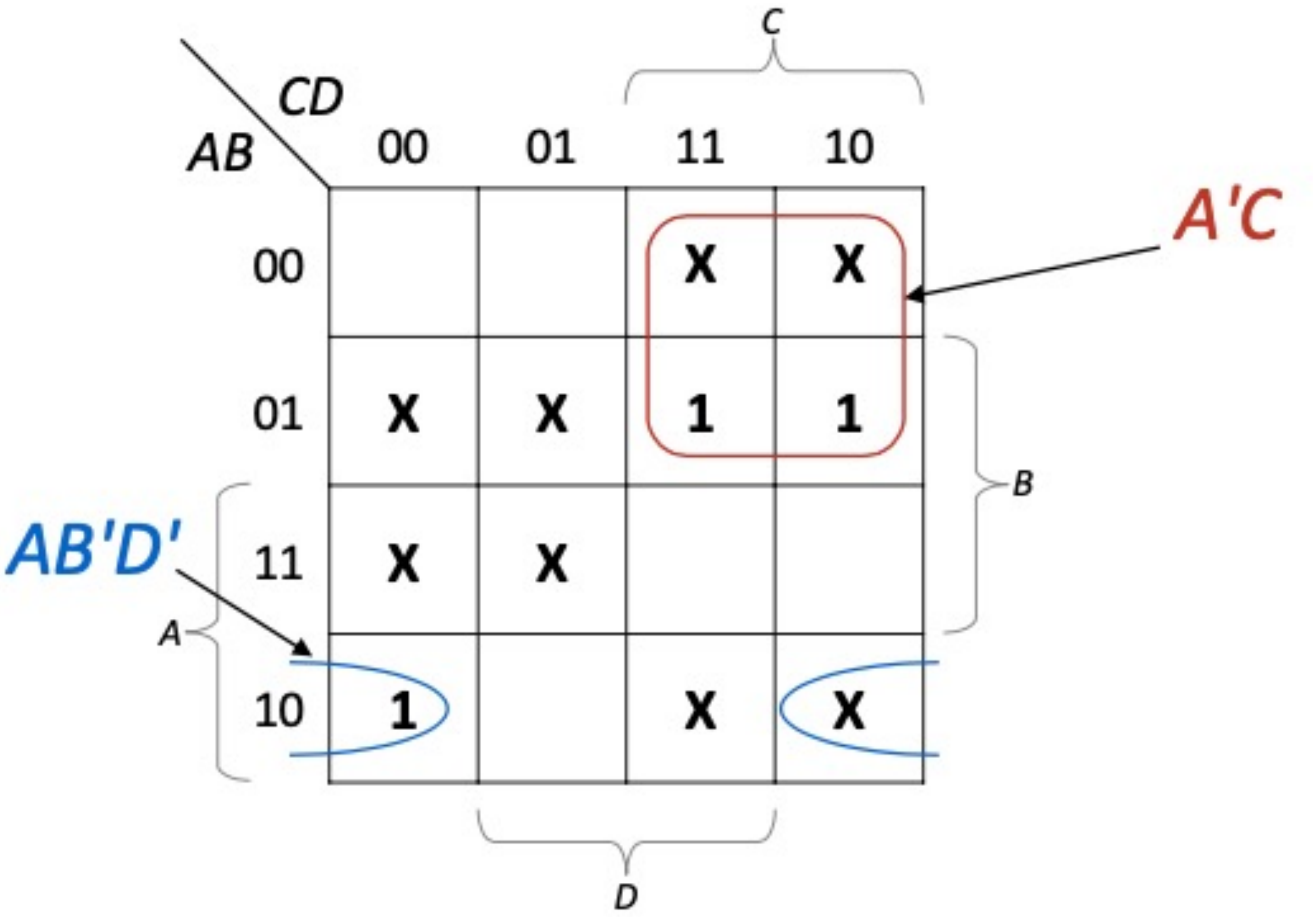
$F1 = BC' + B'C + A'C + AC'D'$



Example cont.

The simplified function has four solutions:

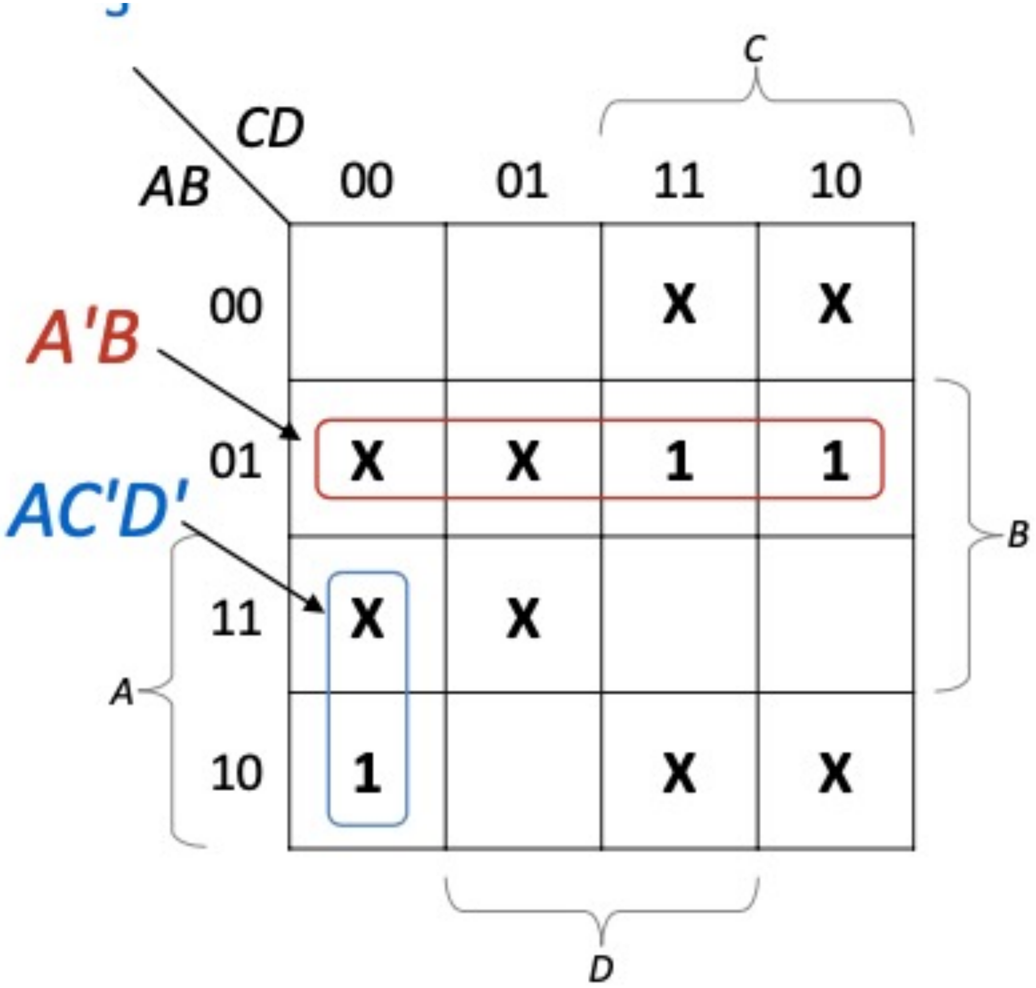
$F2 = BC' + B'C + A'C + AB'D$



Example cont.

The simplified function has four solutions:

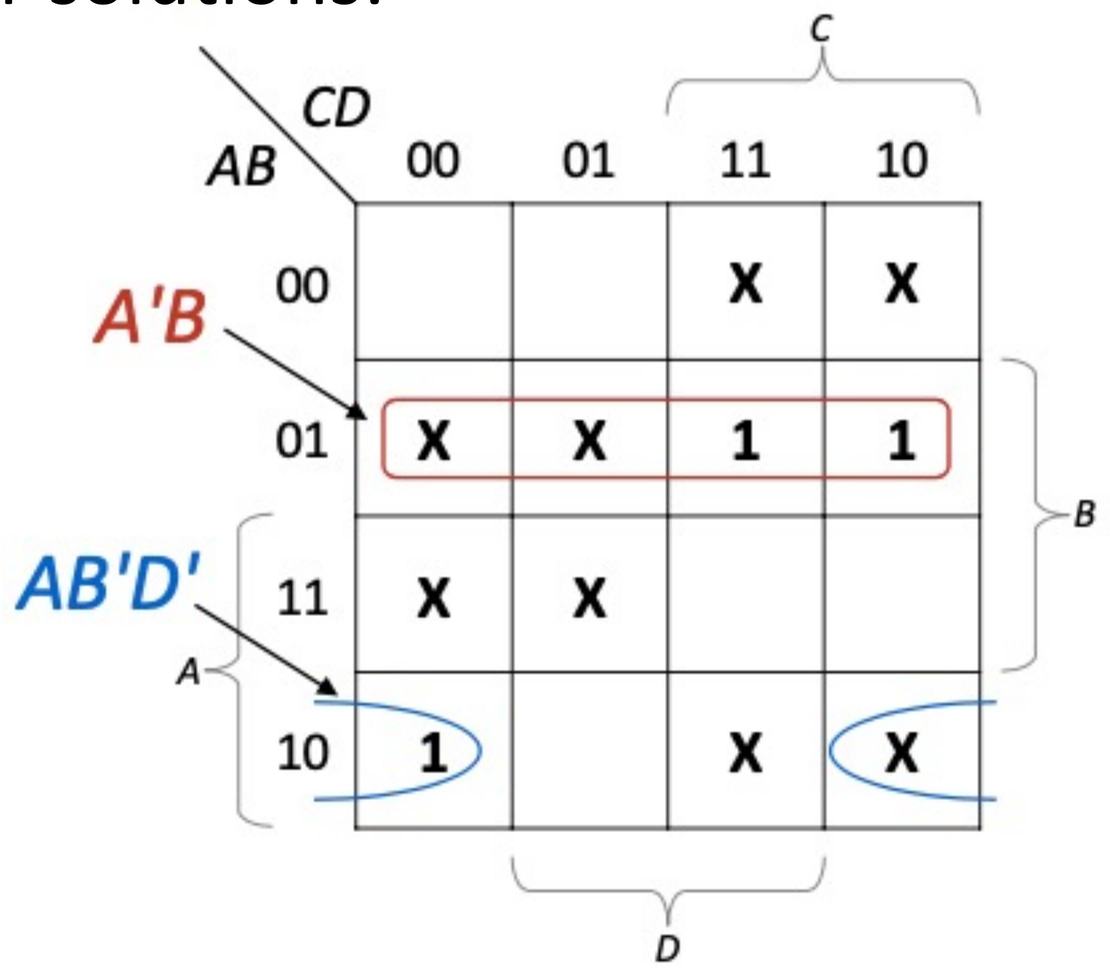
$F3 = BC' + B'C + A'B + AC'D'$



Example cont.

The simplified function has four solutions:

$F4 = BC' + B'C + A'B + AB'D'$



Product Of Sum

Finding a minimum POS expression requires no new theory

The following approach is the simplest:

1. Map complement of function. (If there is already a map for the function, replace all 0’s by 1’s, all 1’s by 0’s, and leave X’s unchanged.)

2. Find the minimum SOP expression for the complement of the function

3. Use DeMorgan’s theorem (P11) to complement that expression, producing POS expression

Example

Consider the Boolean function with minterms and don't cares as:

$$F(A, B, C, D) = \sum m\,(1, 3, 4, 6, 7) + \sum d\,(0, 8, 10, 12, 13)$$

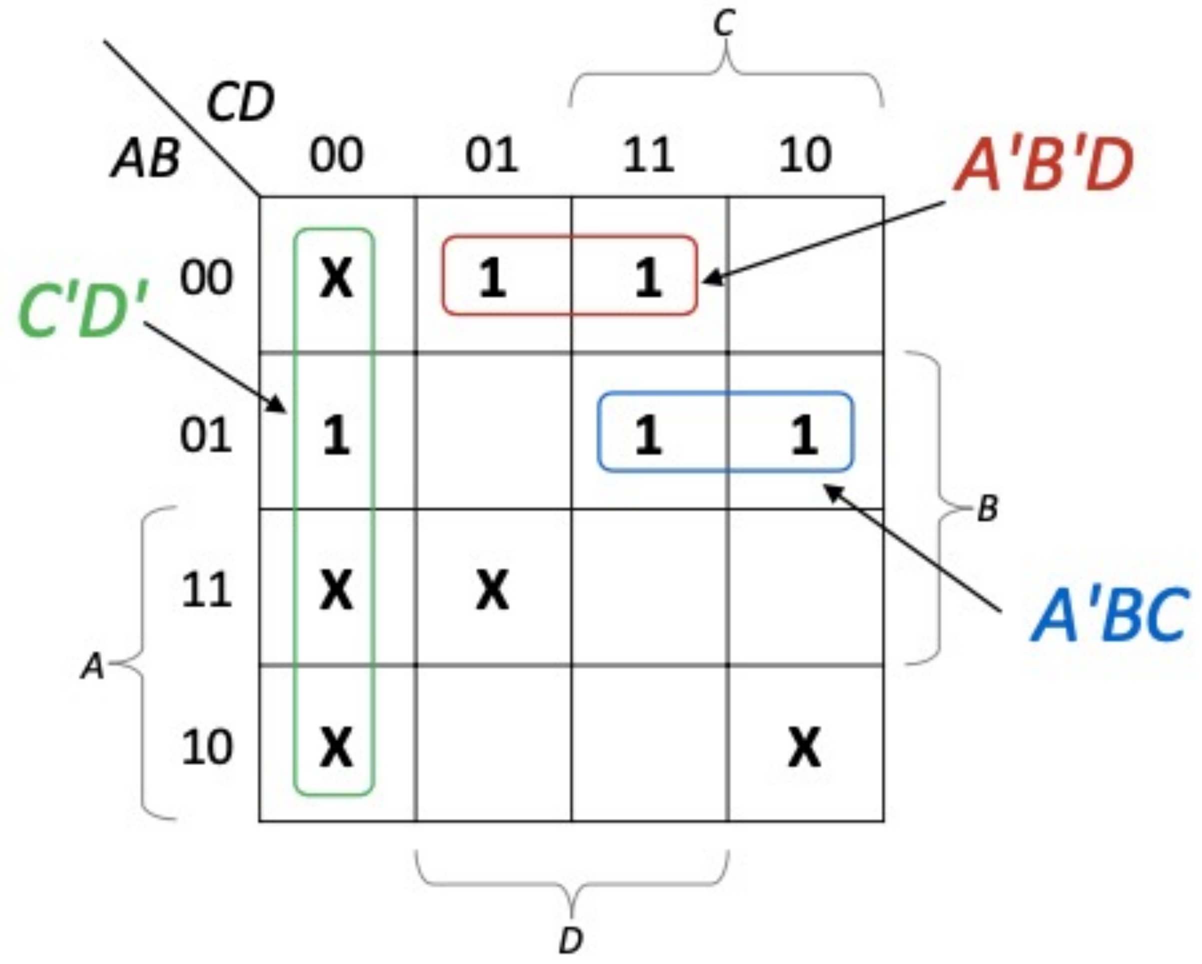
Find minimum SOP and minimum POS for *F*.

- Writing 1's in place of the minterms given in the function and X's in place of the don't cares, we get the following K-Map.

Example cont.

Combining 1's with X's, we get

$F = C'D' + A'B'D + A'BC$ (SOP form)



Example cont.

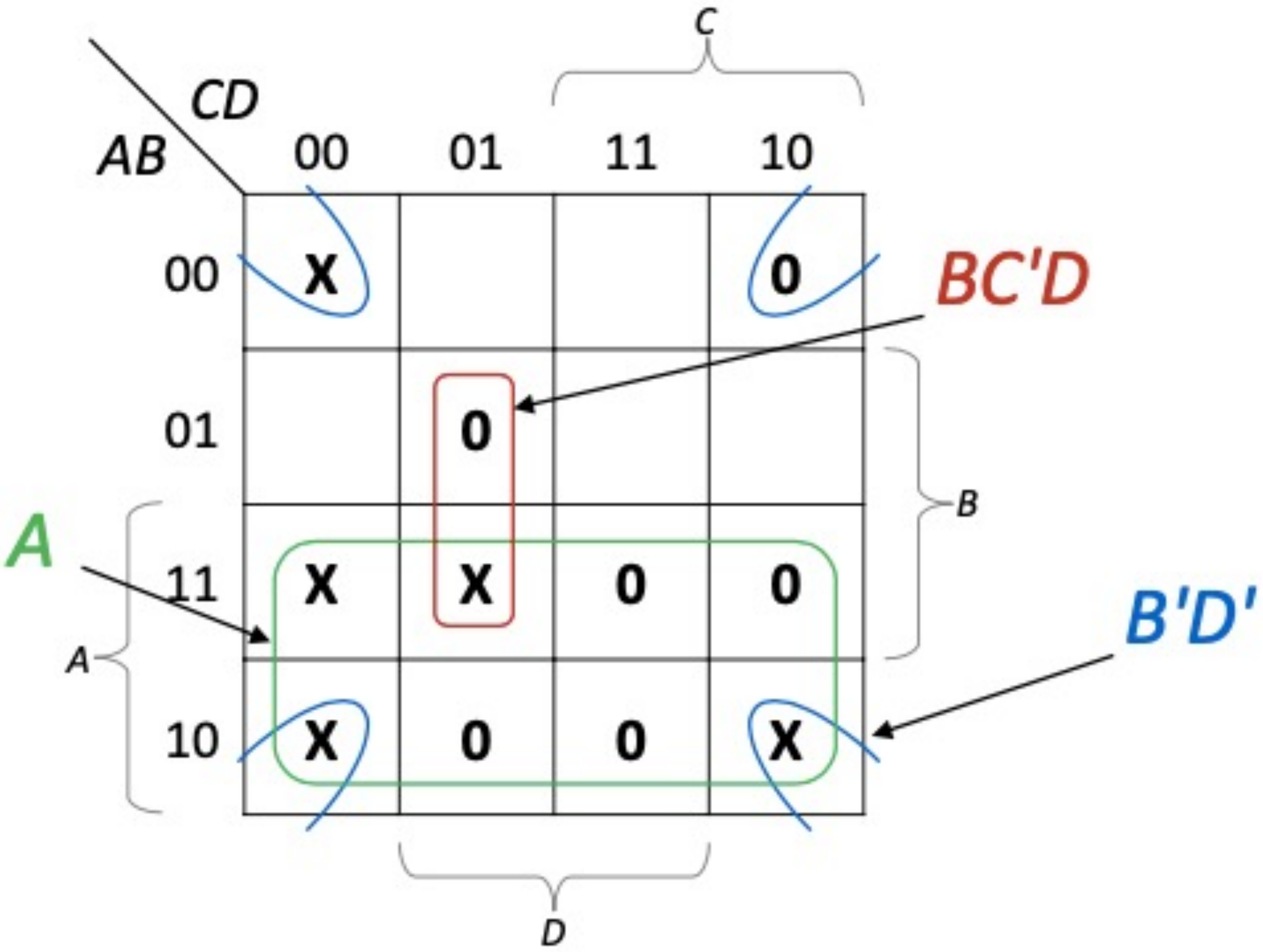
Now filling the remaining boxes by 0's and keeping the X's in their boxes, we get the following K-Map.

AB \ CD	00	01	11	10
00	X			0
01		0		
11	X	X	0	0
10	X	0	0	X

Example cont.

Combining 0's with X's, we get

$F' = A + B'D' + BC'D$



Example cont.

- Taking complement on both sides, we get

$$(F')' = (A + B'D' + BC'D)'$$

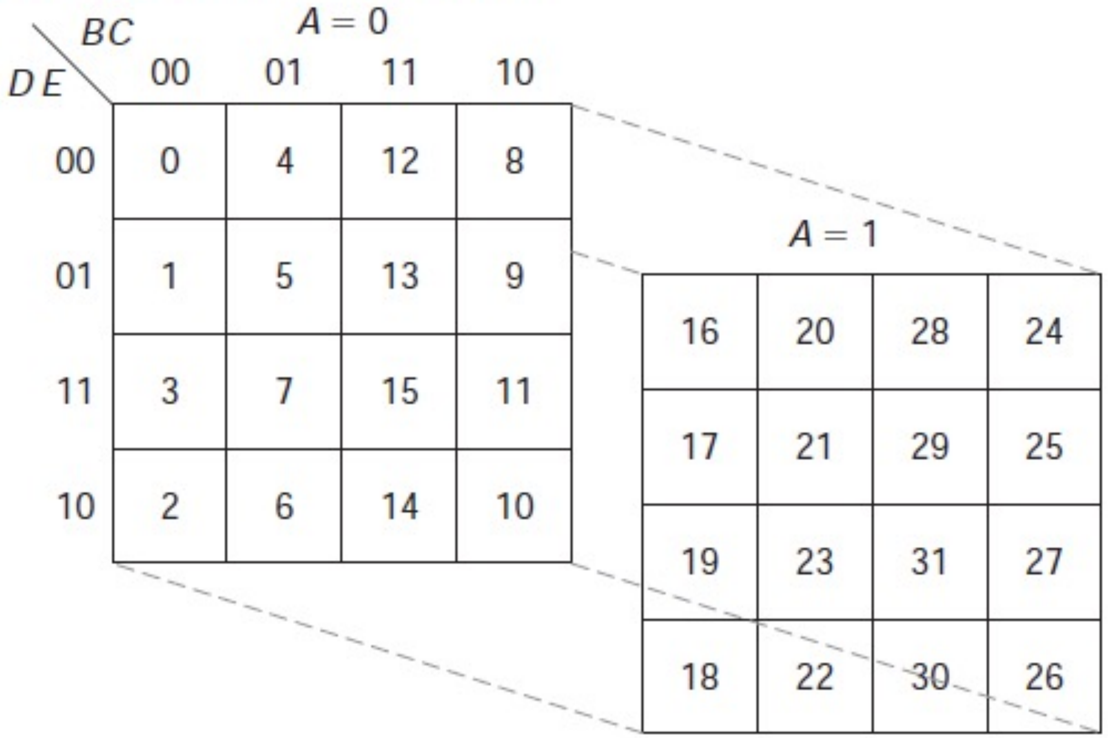
- Applying DeMorgan's law, we get

$$F = A'(B + D)(B' + C + D') \qquad \text{(POS form)}$$

Five and Six Variables Maps

Five variable Karnaugh map contains $2^5 = 32$ squares.

A five-variable Karnaugh Map is shown below



Map 3.16 A five-variable map.

Five Variables Map

Although there are several arrangements that have been used, prefer to look at it as two layers of 16 squares each

- Top layer contains squares for 1st 16 minterms (for which 1st variable, *A*,is 0)
- Bottom layer contains remaining 16 squares
- Each square in bottom layer corresponds to minterm numbered 16 more than

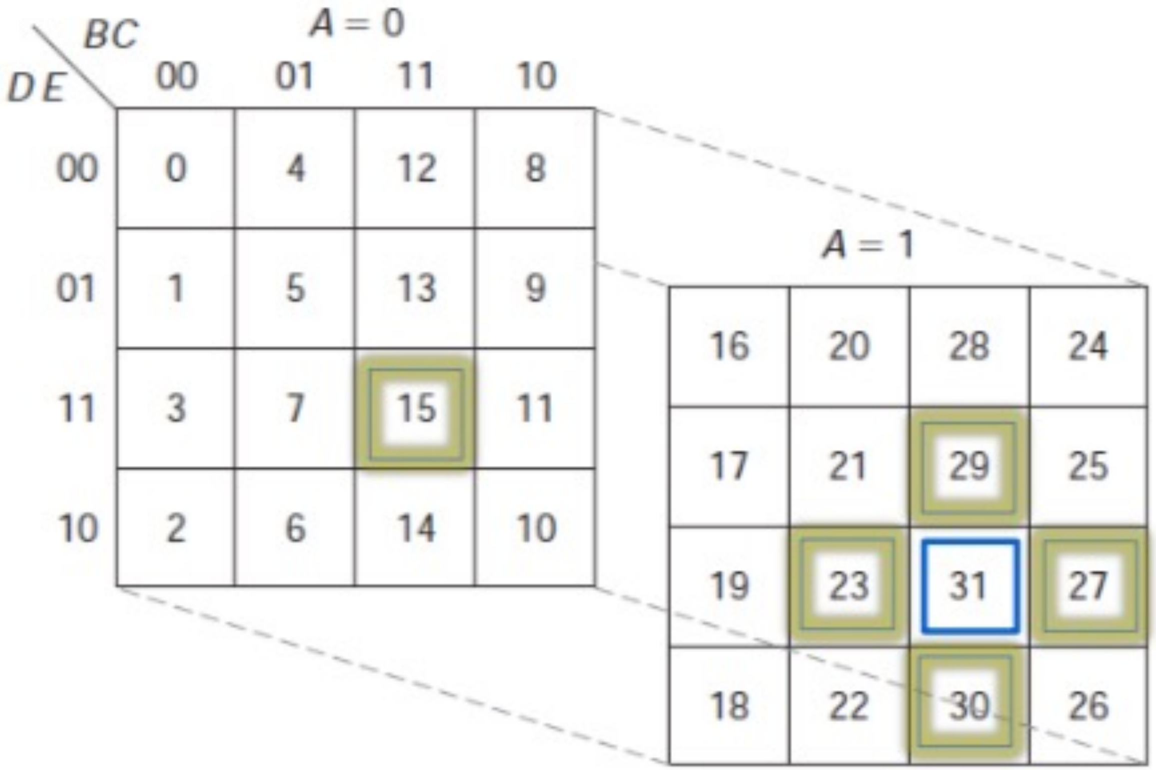
square above it

- Product terms appear as rectangular solids of 1, 2, 4, 8, 16, . . . 1's or X's
- Squares directly above and below each other are adjacent

Five Variables Map

A five variable K-Map must be thought to consist of two four variable maps, which can be recognized from bold lines in the center of the map

– Minterm m_{31} in five variable map is adjacent to m_{30}, m_{15}, m_{29}, m_{23} and m_{27}

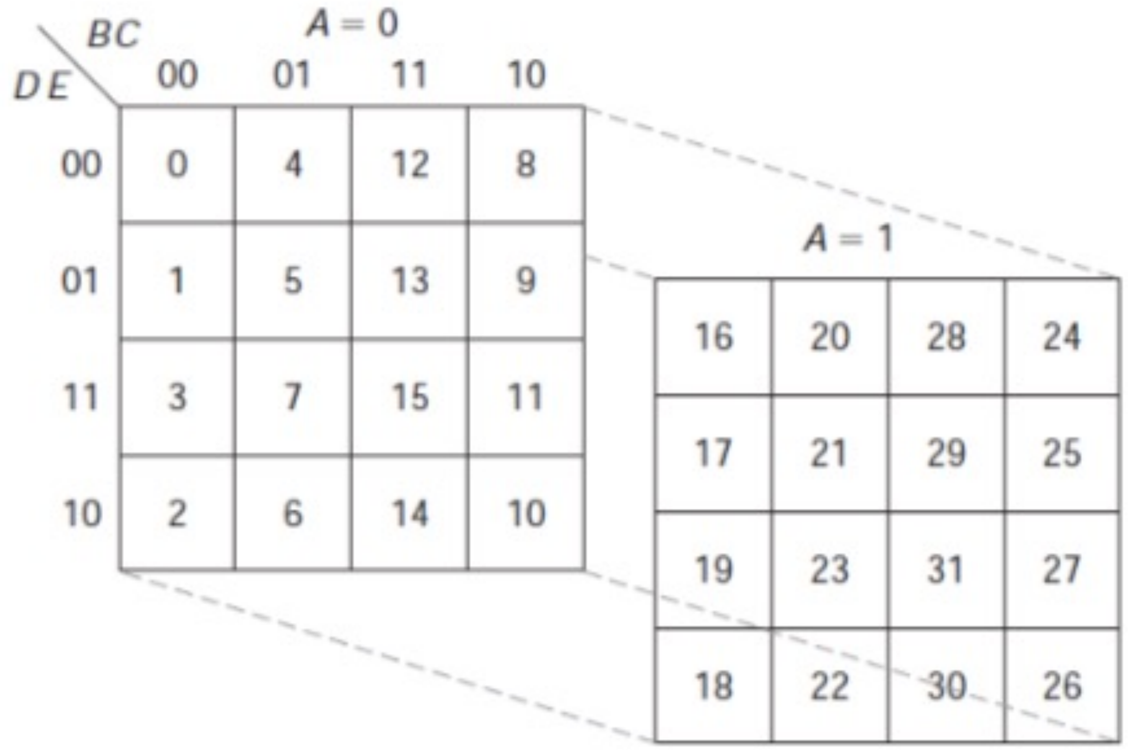


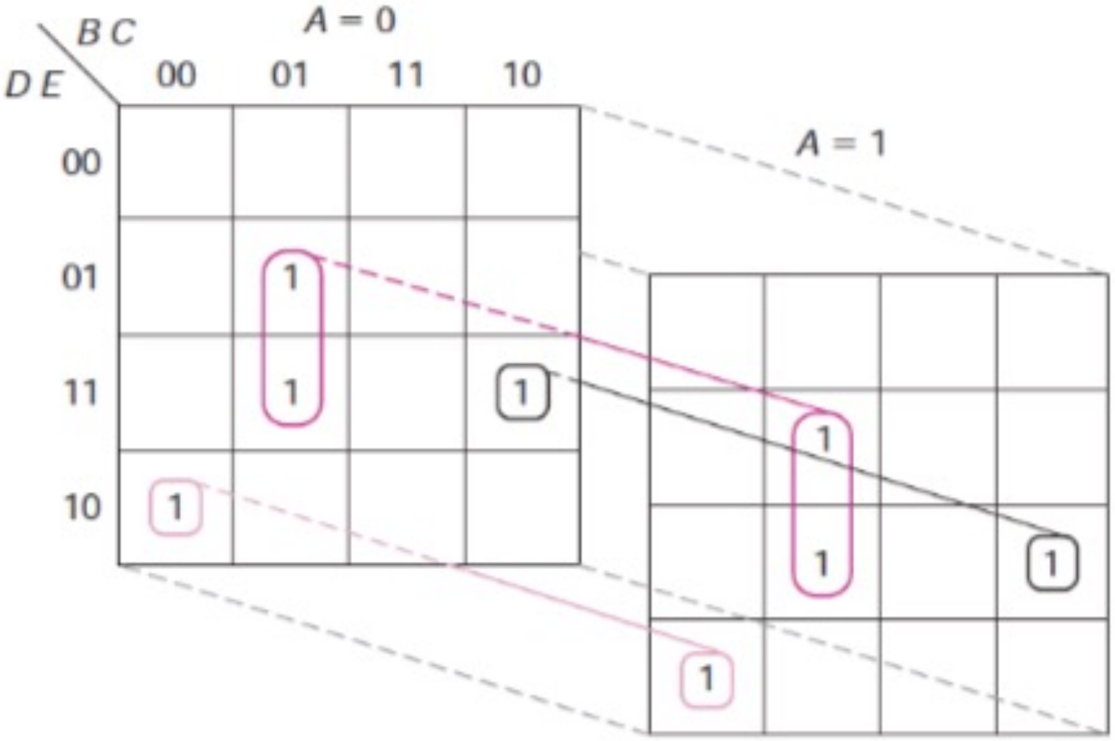
Example

$$m_2 + m_5 = A'B'C'DE' + AB'C'DE' = B'C'DE'$$

$$m_{11} + m_{27} = A'BC'DE + ABC'DE = BC'DE$$

$$m_5 + m_7 + m_{21} + m_{23} = B'CE$$





Example

- $F(A, B, C, D, E) = \Sigma m(4, 5, 6, 7, 9, 11, 13, 15, 16, 18, 27, 28, 31)$

$$\boxed{A'B'C + A'BE + BDE + AB'C'E' + ABCD'E'}$$

A

$A = 0$

BC \ DE	00	01	11	10
00		1		
01	1	1	1	1
11		1	1	1
10		1	1	

$A = 1$

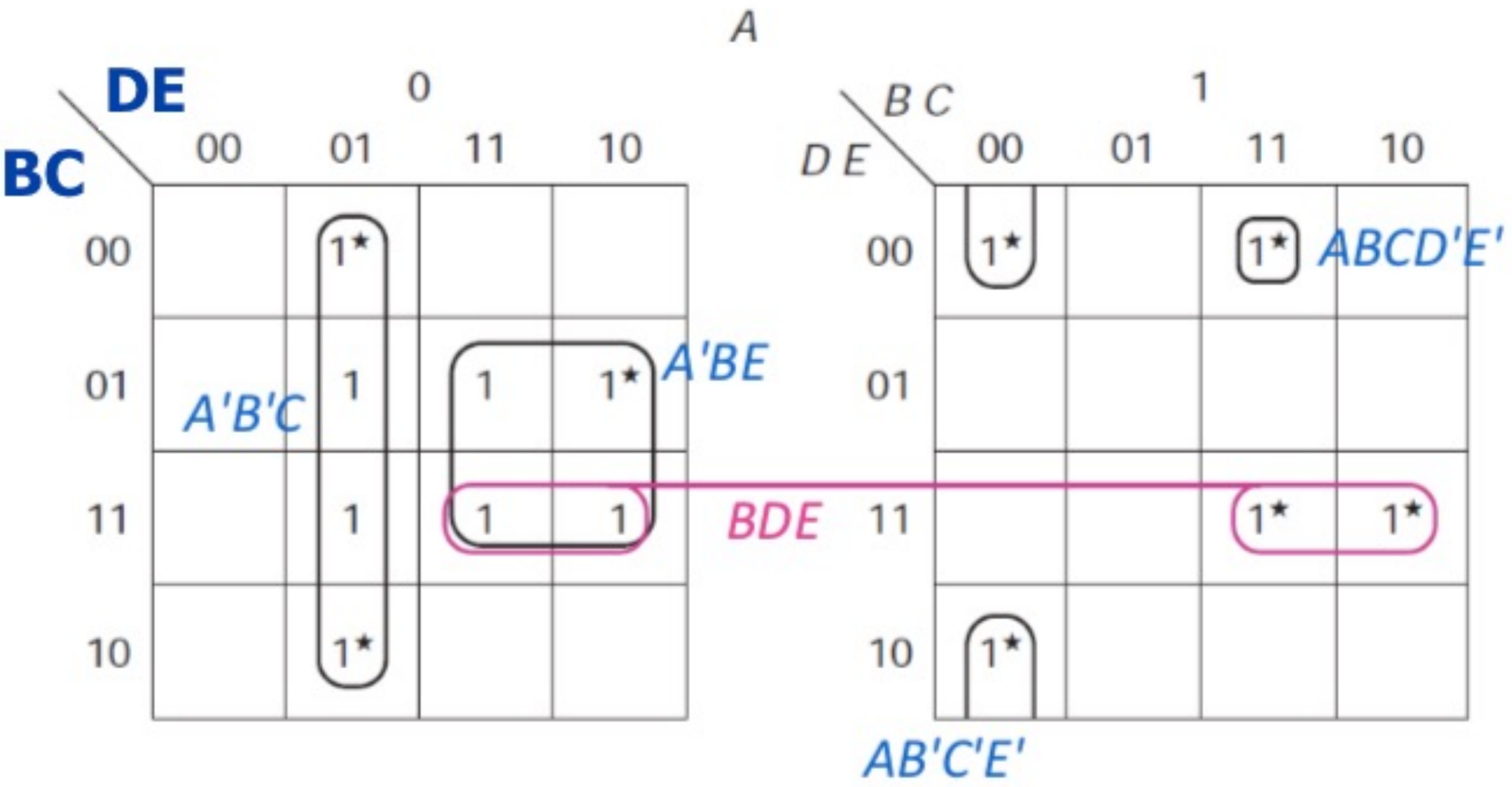
BC \ DE	00	01	11	10
00	1		1	1
01				
11	1		1	1
10	1		1	

Example cont.

A'B'C + A'BE + BDE + AB'C'E' + ABCD'E'

- Simplified Boolean function in SOP form

$$F = A'B'C + A'BE + BDE + AB'C'E' + ABCD'E'$$



Another Example

$$G(A, B, C, D, E) = \sum m(1, 3, 8, 9, 11, 12, 14, 17, 19, 20, 22, 24, 25, 27)$$

A = 0

DE \ BC	00	01	11	10
00			1	1
01	1			1
11	1			1
10			1	

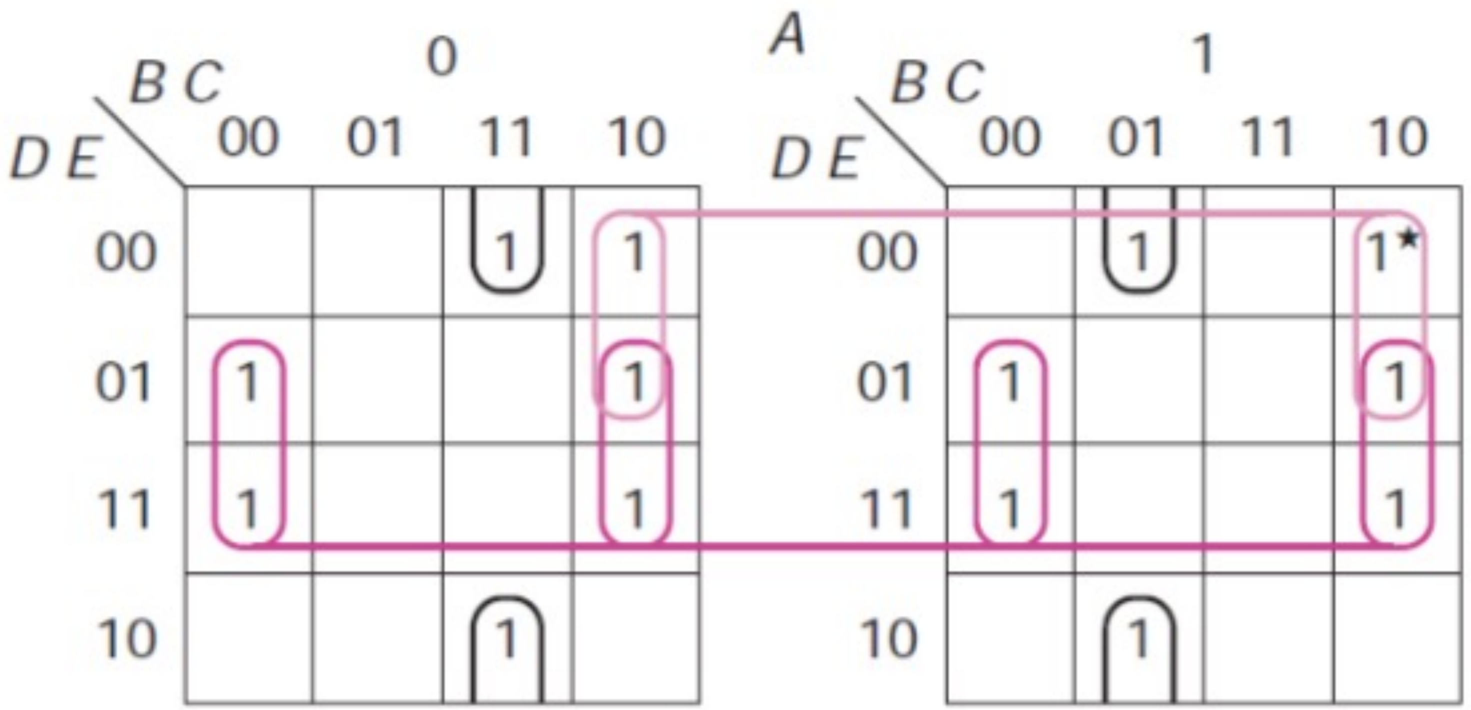
A = 1

DE \ BC	00	01	11	10
00		1		1
01	1			1
11	1			1
10		1		

Example cont.

Combining 1’s, we get the following simplified Boolean function in SOP form as

$$G = C'E + BC'D' + AB'CE' + A'BCE'$$



One More Example

$H(A, B, C, D, E) = \sum m(1, 8, 9, 12, 13, 14, 16, 18, 19, 22, 23, 24, 30)$
$+ \sum d(2, 3, 5, 6, 7, 17, 25, 26)$

- K-map:

$A = 0$

DE \ BC	00	01	11	10
00			1	1
01	1	X	1	1
11	X	X		
10	X	X	1	

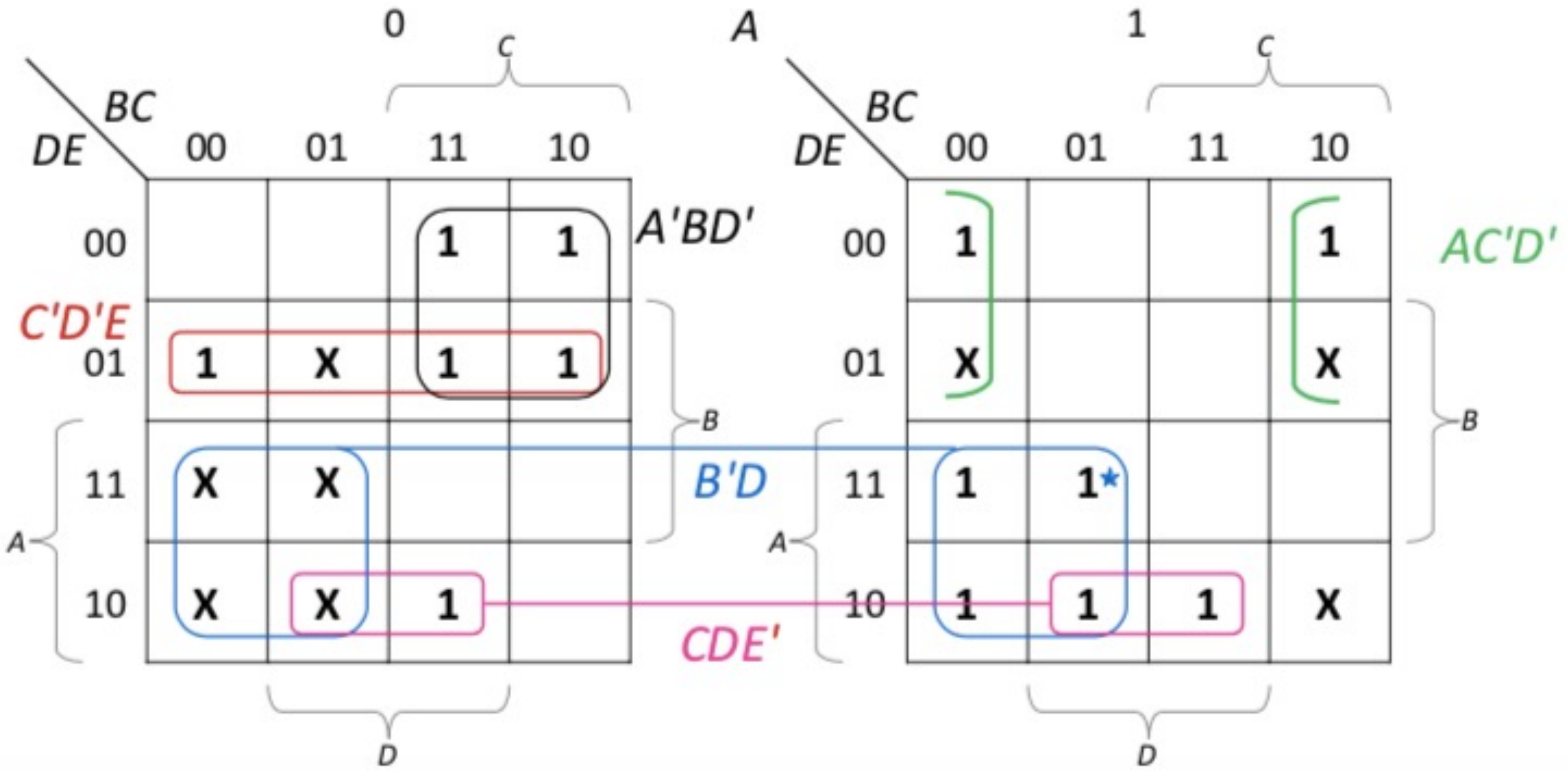
$A = 1$

DE \ BC	00	01	11	10
00	1			1
01	X			X
11	1	1*		
10	1	1	1	X

Example cont.

Combining 1’s, we get simplified Boolean function in SOP form as

$$H = AC'D' + C'D'E + CDE' + A'BD' + B'D$$



Example cont.

We can also get other solutions for this function

All possible eight solutions for this function are as follows:

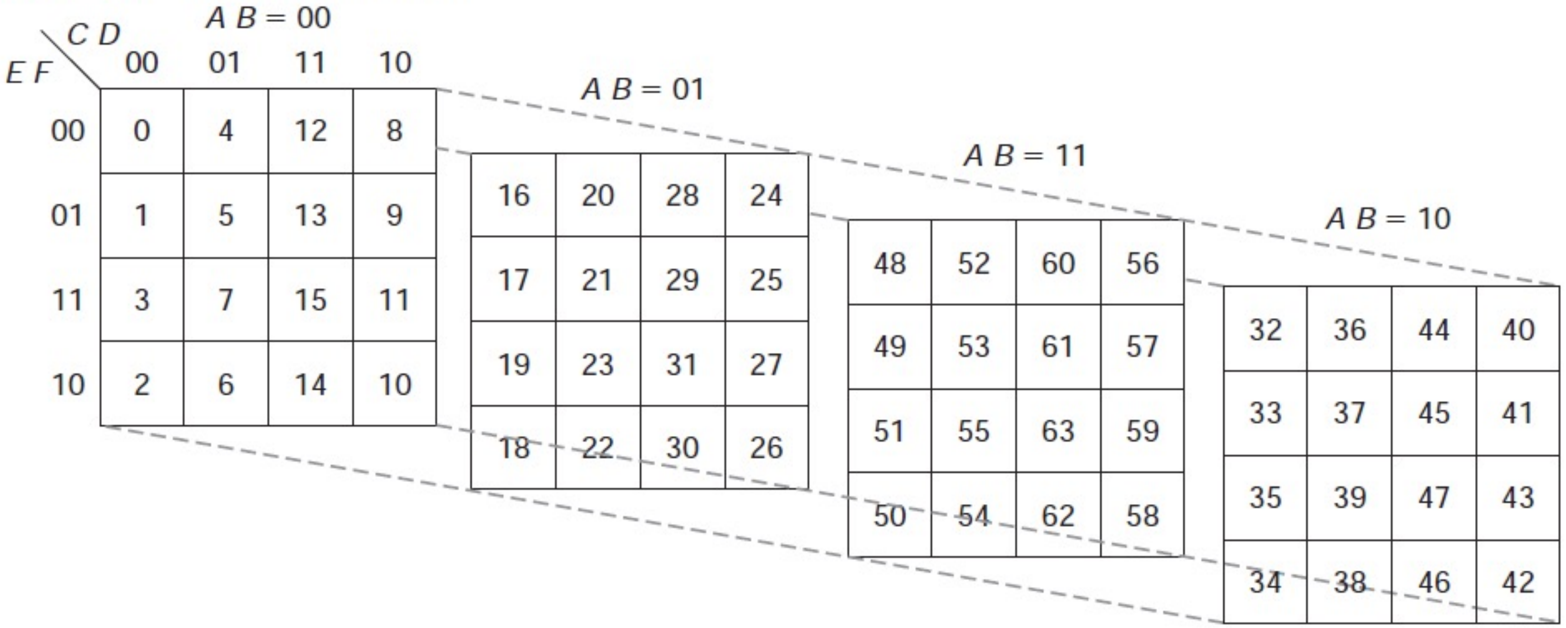
$$H = B'D + CDE' + A'BD' + \begin{Bmatrix} AC'E' \\ AC'D' \end{Bmatrix} + \begin{Bmatrix} A'D'E \\ A'B'E \\ B'C'E \\ C'D'E \end{Bmatrix}$$

Six Variables Map

Contains $2^6 = 64$ squares, drawn as four layers of 16-square maps

- First two variables determine layer
- Other variables specify square within layer
- Layers are ordered in same way as rows and columns, that is 00, 01, 11, 10

Map 3.17 A six-variable map.



Multiple Output Problems

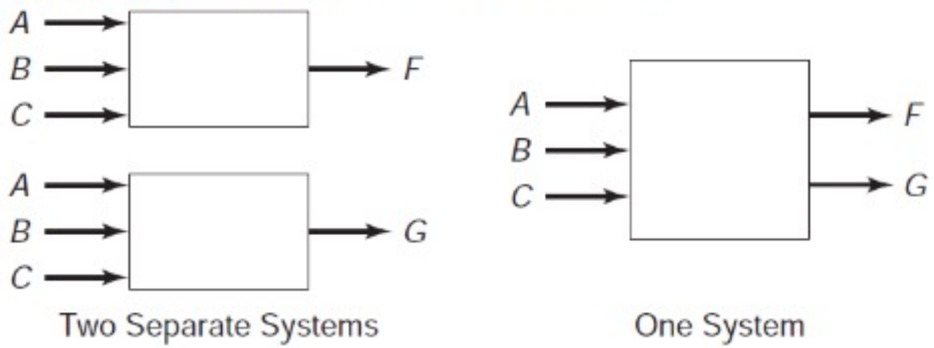
Many problems involve designing system with more than one output

– e.g., a problem with 3 inputs, *A*, *B*, *and C and* 2 outputs, *F and G, could be treated as two separate problems*

– Would then map each of the functions, and find minimum solutions

– However, if this treated as a single system with three inputs and two outputs, may be able to economize by sharing gates

Figure 3.1 Implementation of two functions.



Example

Consider functions:

$F(A, B, C) = \sum m(0, 2, 6, 7)$ $G(A, B, C) = \sum m(1, 3, 6, 7)$

If we map each of these and solve them separately

C \ AB	00	01	11	10
0	1	1	1	
1			1	

F

C \ AB	00	01	11	10
0			1	
1	1	1	1	

G

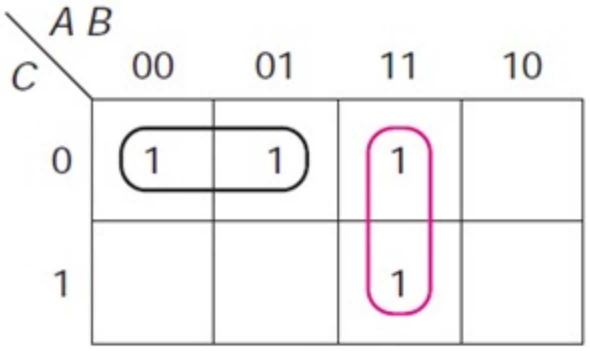
Example cont.

Consider functions:

$F(A, B, C) = \sum m(0, 2, 6, 7)$ $G(A, B, C) = \sum m(1, 3, 6, 7)$

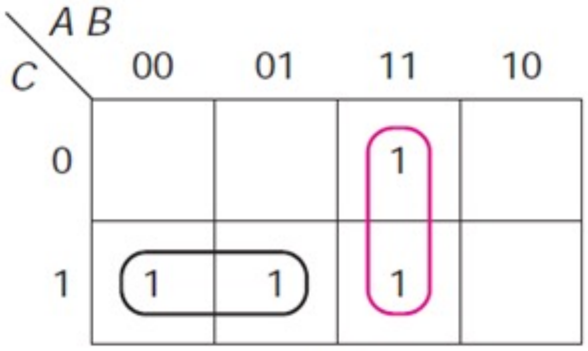
If we map each of these and solve them separately

we obtain:



F

$F = A'C' + AB$



G

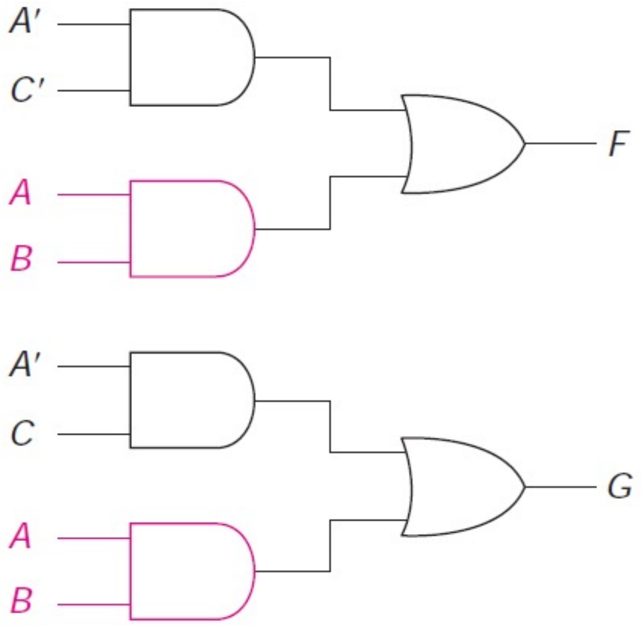
$G = A'C + AB$

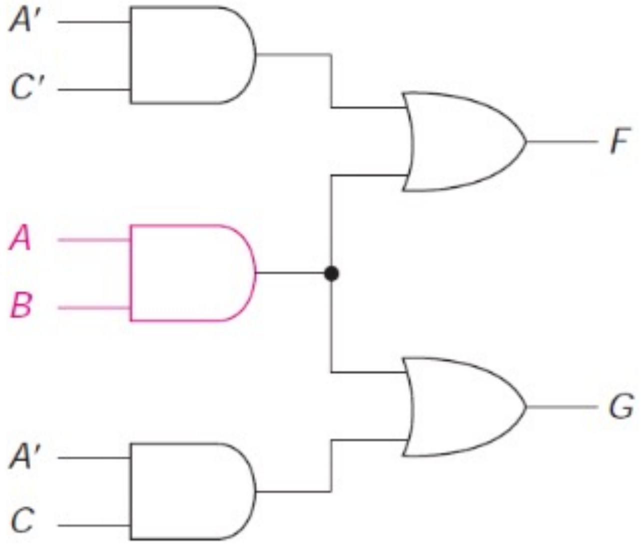
Example cont.

- Looking at maps, we see that the same term *AB* is circled on both
- Thus, can build circuit on the right, rather than two circuits on the left

$F = A'C' + AB$

$G = A'C + AB$





Example

$$F(A, B, C) = \sum m(0, 1, 6) \qquad G(A, B, C) = \sum m(2, 3, 6)$$

C \ AB	00	01	11	10
0	1		1	
1	1			

F

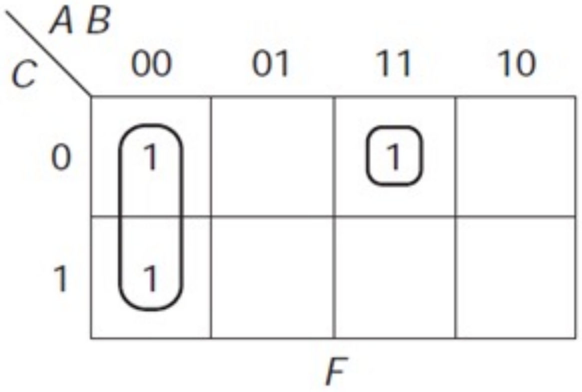
C \ AB	00	01	11	10
0		1	1	
1		1		

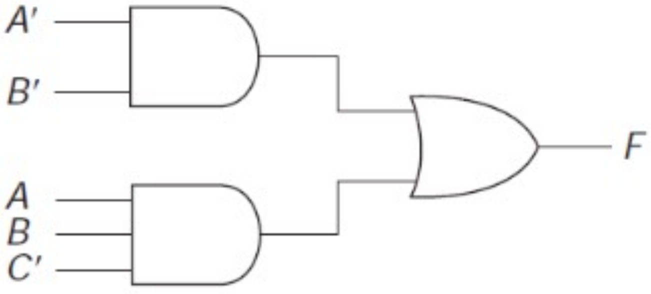
G

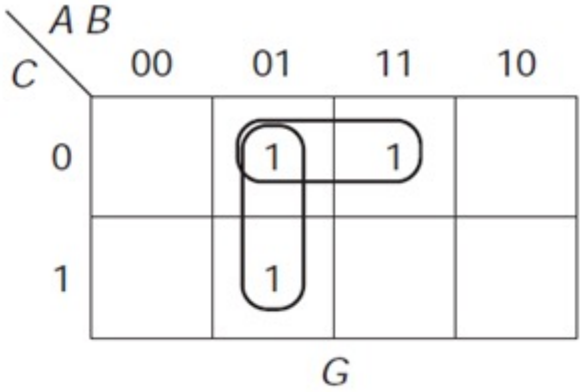
Example cont.

$F(A, B, C) = \sum m(0, 1, 6)$ $\quad$ $G(A, B, C) = \sum m(2, 3, 6)$

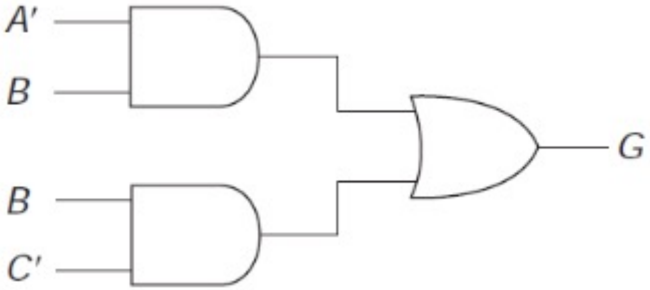
$F = A'B' +$







$G = A'B + BC'$

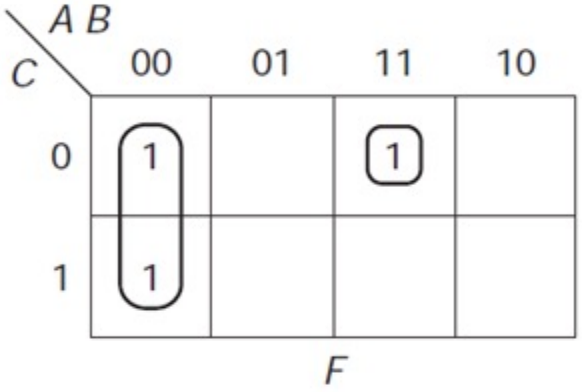


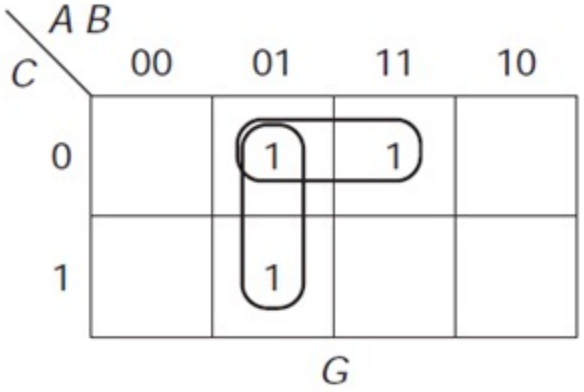
Solution requires 6 gates (4 ANDs and 2 ORs) with 13 inputs

Example cont.

$F(A, B, C) = \sum m(0, 1, 6)$ $G(A, B, C) = \sum m(2, 3, 6)$

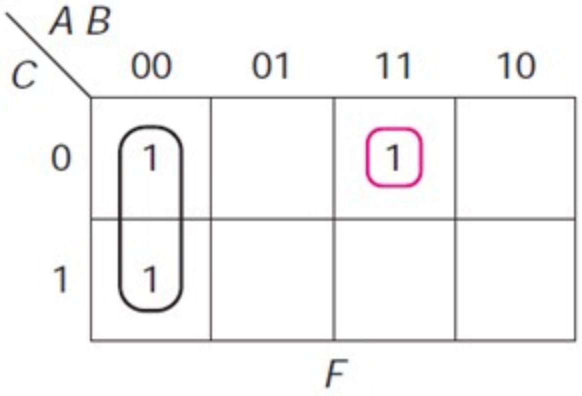
$F = A'B' + ABC'$

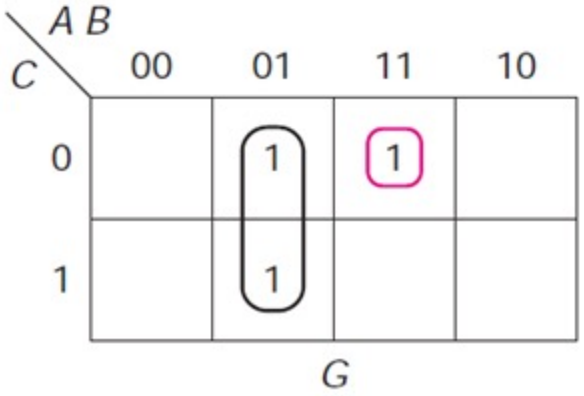




$G = A'B + BC'$

$F = A'B' + ABC'$

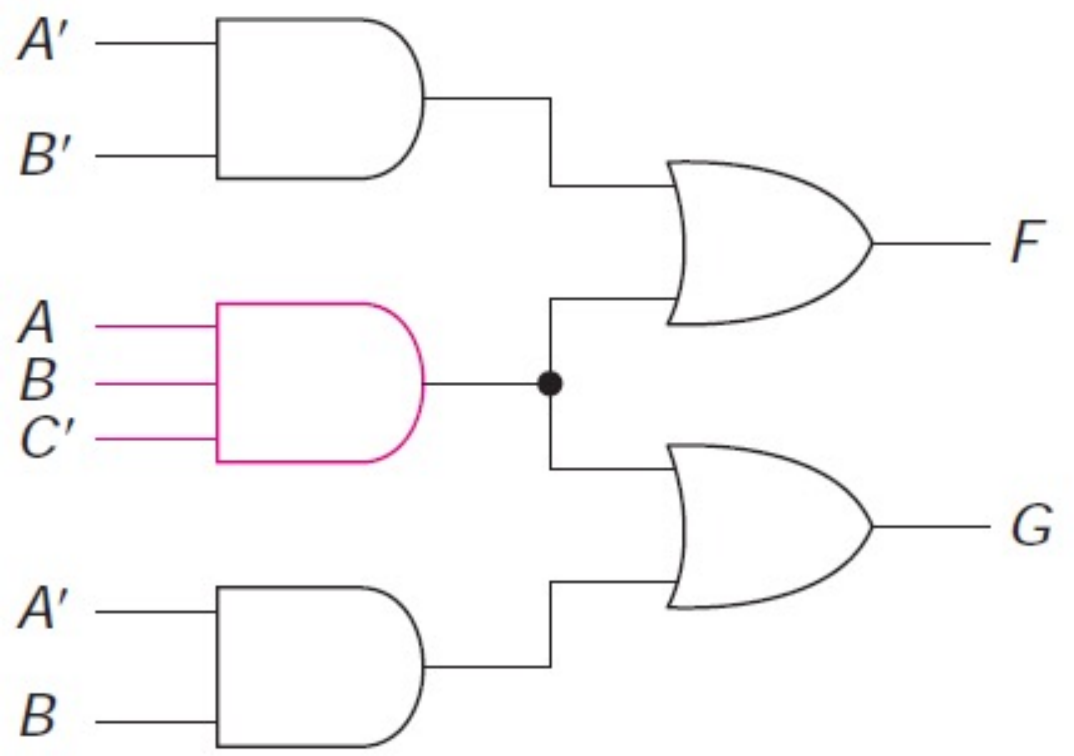




$G = A'B + ABC'$

Example cont.

However, as can be seen from the second pair of maps, we can share the term *ABC* and obtain: $F = A'B' + ABC'$ $G = A'B + ABC'$



Solution requires 5 gates with 11 inputs (compared to: 6 gates & 13 inputs)

Example cont.

- This example illustrates that a shared term in a minimum solution need not be a prime implicant

- For example, here ABC' is a prime implicant of F but not of G

Another Example

- $F(A, B, C) = \Sigma m(2, 3, 7)$ $G(A, B, C) = \Sigma m(4, 5, 7)$

C \ AB	00	01	11	10
0		1		
1		1	1	

F

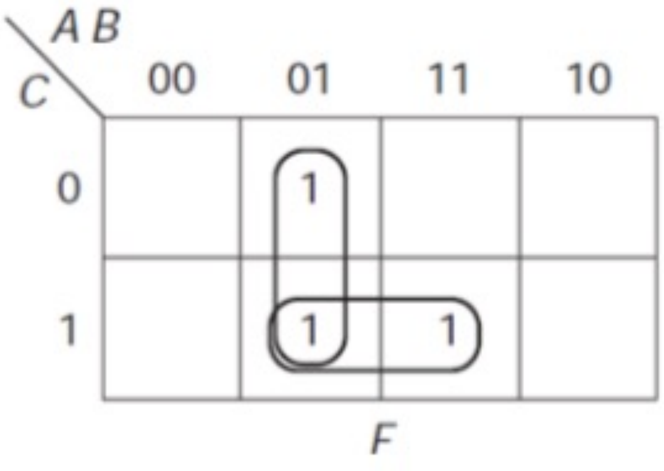
C \ AB	00	01	11	10
0				1
1			1	1

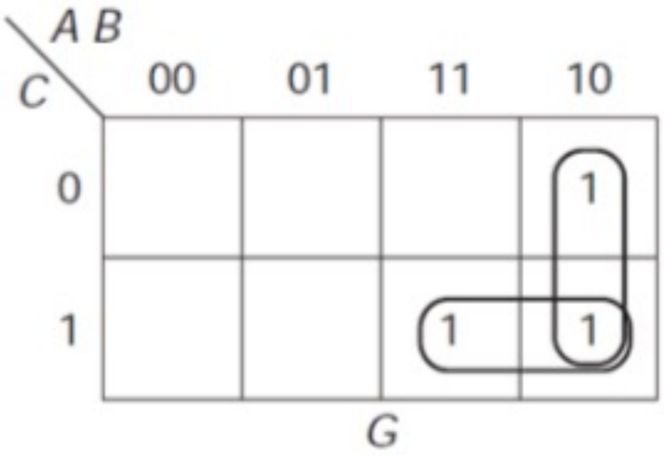
G

Example cont.

$F(A, B, C) = \Sigma m(2, 3, 7)$ $G(A, B, C) = \Sigma m(4, 5, 7)$

$F = A'B + BC$





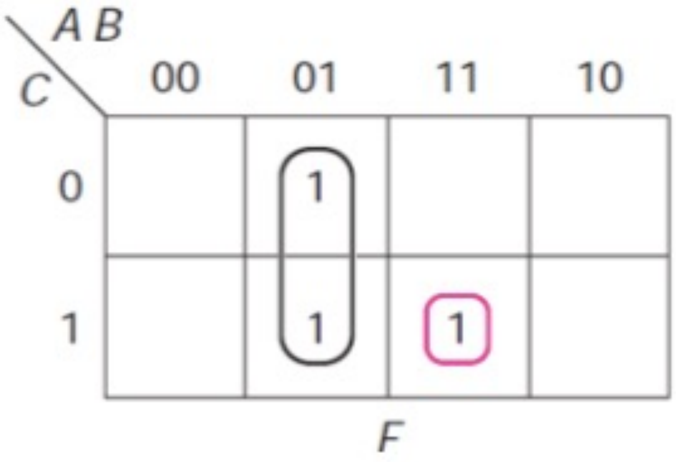
$G = AB' + AC$

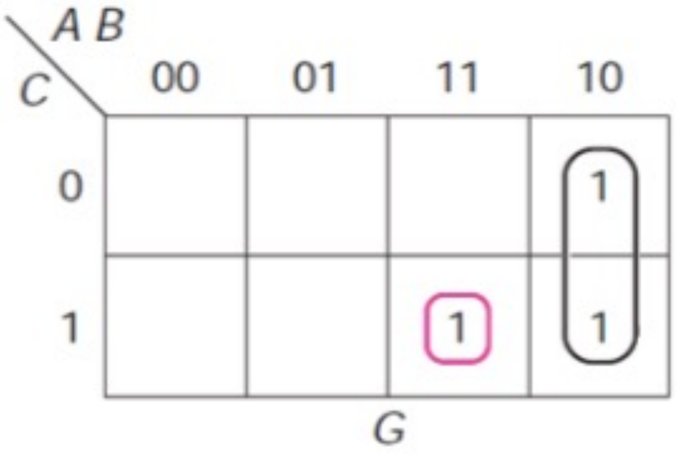
Example cont.

$F(A, B, C) = \Sigma m(2, 3, 7)$

$G(A, B, C) = \Sigma m(4, 5, 7)$

$F = A'B + ABC$

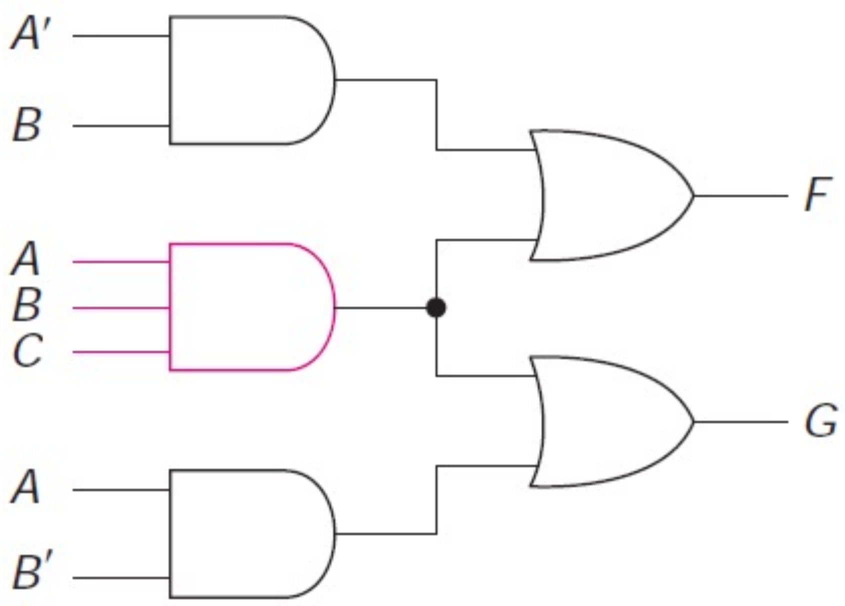




$G = AB' + ABC$

Example cont.

However, as can be seen in the second set of maps, we can share the term *ABC*, even though it is not a prime implicant of either function, a solution that requires only five gates



$F = A'B + ABC$

$G = AB' + ABC$

One More Example

$F(A,B,C,D)=\sum m(4,5,6,8,12,13)$ $G(A,B,C,D)=\sum m(0,2,5,6,7,13,14,15)$

CD \ AB	00	01	11	10
00		1	1	1*
01		1	1	
11				
10		1		

F

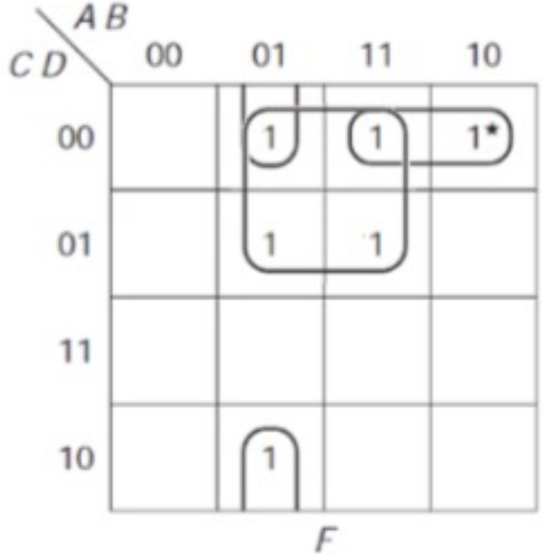
CD \ AB	00	01	11	10
00	1*			
01		1	1	
11		1	1	
10	1	1	1*	

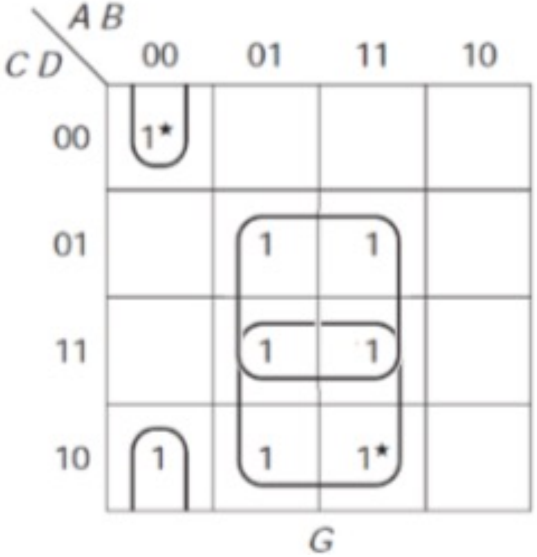
G

Example cont.

$F(A,B,C,D)=\sum m(4,5,6,8,12,13)$ $G(A,B,C,D)=\sum m(0,2,5,6,7,13,14,15)$

$F = AC'D' + A'BD' + BC'$





$G = A'B'D' + BC + BD$

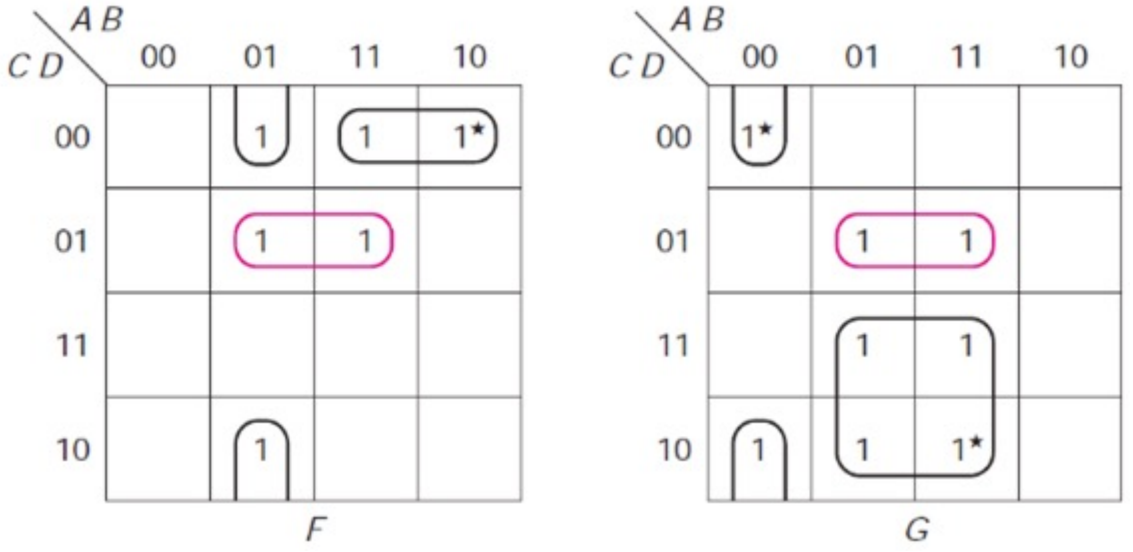
If we had minimized functions individually, we would have used two separate terms for the third term in each expression

Total of 8 gates with 21 gate inputs

Example cont.

Share the term *BC'D*, as shown on the following maps

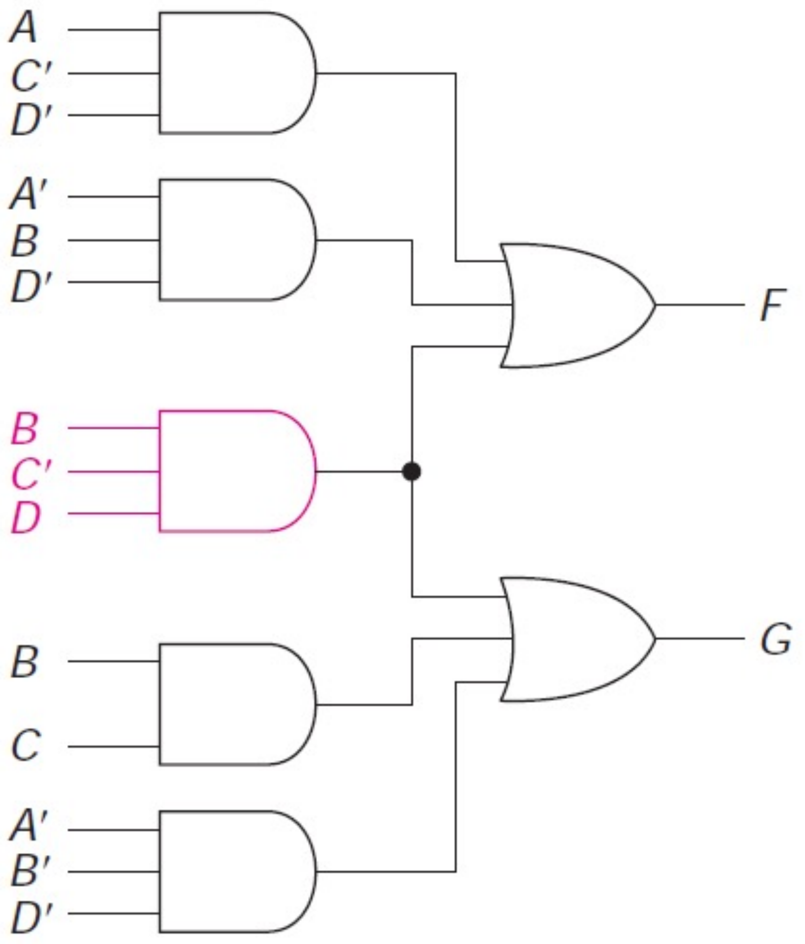
$F = AC'D' + A'BD' + BC'D$



$G = A'B'D' + BC + BC'D$

Implementation will require 7 gates with 20 gate inputs

Example cont.



$F = AC'D' + A'BD' + BC'D$

$G = A'B'D' + BC + BC'D$

The End